

BOUNDED GAPS BETWEEN PRIMES

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ABSTRACT. Polymath 8b [5] proved that $H_1 = \liminf_{n \rightarrow \infty} (p_{n+1} - p_n) \leq 246$. In this paper we show how the Bombieri-Vinogradov theorem can be combined with newer equidistribution estimates for smooth moduli to obtain the improved bound $H_1 \leq 240$.

1. INTRODUCTION

In this paper we study $H_1 = \liminf_{n \rightarrow \infty} (p_{n+1} - p_n)$, the smallest gap between consecutive primes which appears infinitely often. The twin prime conjecture suggests that $H_1 = 2$, but only since Zhang's breakthrough [7] in 2013 is it known that H_1 is finite. Zhang's bound $H_1 \leq 70000000$ has since been improved several times. In particular, Maynard [3] showed that $H_1 \leq 600$, and the Polymath project [5] improved this to the current best known bound $H_1 \leq 246$. A core ingredient of Zhang's result were his newly proved equidistribution estimates for primes in arithmetic progressions with smooth moduli. However, Maynard's new sieve weights later allowed for the use of the older and simpler Bombieri-Vinogradov theorem [3]. Polymath's bound $H_1 \leq 246$ also only uses the Bombieri-Vinogradov theorem and no newer equidistribution estimates [5].

More generally, one may consider $H_m = \liminf_{n \rightarrow \infty} (p_{n+m} - p_n)$, the gap between $m + 1$ consecutive primes. Maynard [3] first showed that H_m is bounded, giving estimate $H_m \ll m^3 \exp(4m)$, and Tao obtained a similar bound around the same time. The Polymath project [5] improved this bound to $H_m \ll \exp((4 - \frac{28}{157})m)$. Baker and Irving [1] first introduced Harman's sieve into this problem, giving $H_m \ll \exp(3.815m)$, which we improved to $H_m \ll \exp(3.8075m)$ [6]. Concrete upper bounds for $H_2, \dots, H_5$ are also known, see [5] and [6]. The best known upper bounds on H_m for $m \geq 2$ all use equidistribution estimates of Zhang type.

Equidistribution estimates of Zhang type give an exponent of distribution larger than $\frac{1}{2}$, improving on the exponent in the Bombieri-Vinogradov theorem, but only apply to a limited choice of moduli, like x^δ -smooth numbers. When applying the GPY method with Maynard-Tao sieve weights, a larger exponent of distribution corresponds to a bigger function support in the relevant optimization problem, whereas a restricted choice of moduli also restricts the support. In Polymath's approach to proving upper bounds on H_m [5], the gains from having a larger exponent of distribution outweigh the losses from being restricted to x^δ -smooth numbers when $m \geq 2$, but not when $m = 1$.

The aim of this paper is to combine the function support given by the Bombieri-Vinogradov theorem with that of newer equidistribution estimates, so that we may benefit from the advantages of both, and improve the bound on H_1 . We prove the following:

Theorem 1. *The quantity $H_1 = \liminf_{n \rightarrow \infty} (p_{n+1} - p_n)$ satisfies $H_1 \leq 240$.*

The value 246 corresponds to the length of the shortest admissible 50-tuple, while 240 is the length of the shortest admissible 49-tuple. The bound $H_1 \leq 240$ is therefore the smallest possible improvement. However, while Polymath [5] used symmetric polynomials of degree ≤ 27 to achieve $H_1 \leq 246$, we only needed polynomials of degree ≤ 21 to get $H_1 \leq 240$. As such, we believe that our method could achieve bigger improvements, but due to limited resources and limited time, we have decided to present this modest improvement as a proof of concept.

1.1. Proof strategy. By using equidistribution estimates for moduli with a large smooth factor, we can choose a considerably larger support for the functions used in the GPY optimization problem, giving better results in the corresponding eigenvalue computation. Our strategy is outlined below.

First we need to determine a suitable function support. On the one hand, this support should capture many tuples to which the available equidistribution estimates are applicable, but on the other hand it should be simple enough that we can efficiently integrate over the chosen region. Our basic idea is to

consider a union $S = S_{BV} \cup S_Z$ of the following sets:

$$S_{BV} = \left\{ (t_1, \dots, t_k) \in [0, 1]^k : t_1 + \dots + t_k \leq \frac{1}{4} \right\},$$

$$S_Z(\delta, \omega, \underline{b}) = \left\{ (t_1, \dots, t_k) \in [0, 1]^k : t_1 + \dots + t_k \in \left[\frac{1}{4}, \frac{1}{4} + \omega \right] \text{ and } \sum_{i \in I} t_i \leq b_{|I|} \text{ for } I = \{i : t_i > \delta\} \right\}.$$

To decide whether S is a suitable support, we must ask if we have the necessary equidistribution estimates for all moduli $m = d_1^{t_1} \dots d_k^{t_k} e_1^{s_1} \dots e_k^{s_k}$ with $((t_1, \dots, t_k), (s_1, \dots, s_k)) \in S \times S$. For elements of $S_{BV} \times S_{BV}$ we can simply apply the Bombieri-Vinogradov theorem. On the other hand, the condition $\sum_{i \in I} t_i \leq b_{|I|}$ ensures that elements of $(S \times S) \setminus (S_{BV} \times S_{BV})$ have a large x^δ -smooth factor. This x^δ -smooth factor is convenient for applying equidistribution estimates, provided we choose the bounds $\underline{b} = (b_1, b_2, \dots)$ suitably small. (To increase the support further, we will also apply the epsilon-enlargement trick of Polymath [5]. Additionally, to restrict the choice of sequences for which equidistribution estimates must apply, we will implement Harman's sieve.)

However, while the moduli corresponding to $(S \times S) \setminus (S_{BV} \times S_{BV})$ have a reasonably large x^δ -smooth factor, they are not necessarily x^δ -smooth themselves. In particular, moduli coming from $S_{BV} \times S_Z$ may have a prime factor of size up to $x^{1/4}$, meaning they are not even i -tuply x^δ -densely divisible and thus neither the equidistribution estimates of Polymath [4] nor those proved by the author in [6] are directly applicable to the chosen support. As such, we must prove variations of the equidistribution estimates of [4] and [6] in which the restrictions on the moduli have been considerably relaxed.

Having chosen a support and proved suitable equidistribution estimates, we need to solve the optimization problem produced by the GPY method with Maynard-Tao sieve weights. This involves maximizing a ratio of integrals, and can be transformed into an eigenvalue computation for a matrix whose entries are determined by these integrals. Here one needs to choose a basis of functions to integrate. To prove $H_1 \leq 246$, Polymath [5] used symmetric polynomials of the form $\cdot p(t_1, \dots, t_k)^2 (1 - t_1 - \dots - t_k)^b$, where $p(t_1, \dots, t_k)$ is itself a symmetric polynomial of degree a and where $2a + b \leq 27$. To show the strength of our method, we will use the same kind of basis to prove $H_1 \leq 240$, but with the stricter condition $2a + b \leq 21$. This means we get an improved result despite working with a significantly smaller basis.

However, trying to integrate symmetric polynomials over $S = S_{BV} \cup S_Z$ causes significant complications: We need to obtain exact integrals as even minor errors in the matrix entries can quickly lead to incorrect results. However, trying to use Mathematica to integrate a polynomial in 49 variables over a region like this seems pretty hopeless: even for small degree polynomials and 10 variables a computation can already take 10 minutes or more, and the duration appears to increase exponentially, so that we would never get any result for 49 variables. Therefore we need to find a better way to integrate polynomials over S .

Some of the key ideas of our integral computation are as follows: We first consider sets

$$T_s(k) = \{(t_1, \dots, t_k) \in [0, \delta]^k : t_1 + \dots + t_k \leq 1\},$$

$$T_b(k) = \{(t_1, \dots, t_k) \in [\delta, 1]^k : t_1 + \dots + t_k \leq 1\}$$

and polynomials $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$. The integrals of $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$ over $T_s(k)$ and $T_b(k)$ can be expressed as polynomials in δ , with coefficients only dependent on the value of $\lfloor \frac{1}{\delta} \rfloor$. To compute these coefficients we find a recursion which relates an integral over $T_s(k)$ to integrals of polynomials of equal or lower degree over $T_s(k-1)$ and $T_b(k-1)$. We then also integrate over more complicated sets, like for instance $T_m = \{(t_1, \dots, t_r, t_{r+1}, \dots, t_k) \in [\delta, 1]^r \times [0, \delta]^{k-r} : t_1 + \dots + t_k \leq 1\}$. Such integrals can be expressed as matrix multiplications, using the coefficient vectors for $T_s(k)$ and $T_b(k)$ produced in the previous step. The actual integral over $S = S_{BV} \cup S_Z$ is then split up into integrals over scaled T_m , so that we can use the results of the previous steps.

Overall, our calculation of the needed matrix entries involves no integration at all, it only requires matrix multiplication. However, many very large matrices need to be multiplied together, and so despite a lot of effort to speed up our code, it can still take days to compute all necessary matrix entries for symmetric polynomials of degree $2a + b \leq 21$. Our code also uses a lot of memory, a trade off from focusing on computation speed. Due to these time and memory constraints, it was not practical for us to consider symmetric polynomials of degree $2a + b \leq 27$. As such, we believe that with greater resources the same method could be used to get a better bound on H_1 .

1.2. Structure of the paper. In Section 2 we carry out the key steps of the GPY method with Maynard-Tao sieve weights and Polymath's epsilon-enlargement trick. Our arguments follow Polymath's approach [5], but many adjustments need to be made to deal with a more complicated function support and account for the use of Harman's sieve.

In Section 3 we record the equidistribution estimates used in our proof. Although the relevant estimates are closely based on [4] and [6], the moduli restrictions in these papers are too strong, and we need to prove relaxed conditions on the moduli.

The function support used in Section 2 is quite general and involves many different parameters. Dependent on these parameters, the equidistribution estimates of Section 3 may or may not be good enough to cover all relevant moduli. In Section 4 we show how to calculate good parameters for which the available equidistribution estimates work well enough.

In Section 5 we briefly discuss how integrals over the chosen support can be expressed in terms of matrix multiplication and recursion.

In Section 6 we combine the results of the previous sections to complete the proof of Theorem 1.

1.3. Notation. Below is a list of notation and terminology used throughout this paper.

In this paper, lower case Roman letters usually denote integers, and letters p and p_i denote primes.

1_S is the indicator function of set S and $1_{\mathbb{P}}$ is the prime indicator function.

$\mu(n)$ is the Möbius function and $\phi(n)$ is Euler's totient function. $\tau(n)$ is the number of divisors of n .

We say a positive integer n is y -smooth if all its prime factors are of size less than y .

We write $P(y) = \prod_{p < y} p$. If $n \mid P(y)$, then n is y -smooth and squarefree.

We use Vinogradov notation ($\ll$ and $\gg$) and write $X \asymp Y$ if $Y \ll X \ll Y$. If we write $X \ll_A Y$, then the implied constant may depend on A .

2. THE GPY SIEVE

In this section we present a version of the GPY method. The GPY method was first developed by Goldston, Pintz and Yıldırım [2], who proved that $\liminf_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{\log p_n} = 0$. Since then, it has been used in all proofs of upper bounds on H_1 , from Zhang's proof of $H_1 \leq 70000000$ [7] to Polymath's proof of $H_1 \leq 246$ [5]. We will use the GPY sieve weights developed by Maynard in [3], and a modification of Polymath's epsilon enlargement trick with a new function support. Additionally, like Baker and Irving [1], we will use a minorant for the primes, constructed via Harman's sieve.

2.1. Important definitions. Before we state an amended version of the GPY sieve, we need to introduce some definitions. We first define the function support we use to construct Maynard-Tao sieve weights.

Definition 1 (Simplex subset). *Let $k \in \mathbb{N}$, $\delta > 0$ and $\varepsilon > 0$. Let $n \in \mathbb{N}$ and $\underline{A} = (A_0, A_1, \dots, A_n)$ with*

$$-\varepsilon = A_0 < \dots < A_n < \frac{1}{2} - \varepsilon.$$

Let $\underline{B}$ be a $n \times \lfloor \frac{1}{\delta} \rfloor$ matrix with (j, m) th entry equal $B_{j,m}$. Matrix $\underline{B}$ must satisfy that for any $m \geq 1$,

$$\delta < B_{j,m} \leq B_{j,m+1} \leq B_{j,m} + \delta.$$

Then $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ is defined as follows:

$$T_k(\delta, \underline{A}, \underline{B}, \varepsilon) = \bigcup_{j=1}^n \left\{ (t_1, \dots, t_k) \in [0, 1]^k : \sum_{i=1}^k t_i \in [A_{j-1} + \varepsilon, A_j + \varepsilon) \text{ and } \sum_{i \in I} t_i \leq B_{j, |I|} \text{ for } I = \{i : t_i > \delta\} \right\}.$$

Specific values for parameters A_j and $B_{j,m}$ will be chosen at a later stage, once we have derived equidistribution estimates for moduli with large x^δ -smooth factors. The condition $\delta < B_{j,m} \leq B_{j,m+1} \leq B_{j,m} + \delta$ is quite important: If $\sum_{i \in I} t_i \leq B_{j, |I|}$ for $I = \{i : t_i > \delta\}$, this condition implies that also $\sum_{i \in I'} t_i \leq B_{j, |I'|}$ for any $I' \subseteq I$. This property will be used in several proofs in this section.

Next we define a set of corresponding moduli for which we need equidistribution estimates.

Definition 2 (Relevant moduli). *Let $x > 0$ and $\varepsilon_0 > 0$. Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1.*

Further let $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ be the following set of integers in $[1, x]$:

$$\left\{ ee' \prod_{i=1}^m f_i \prod_{i'=1}^{m'} f_{i'}' \in [1, x] : \begin{array}{ll} \log_x(f_1 \dots f_m) \leq (1 - \varepsilon_0)B_{j,m}, & \log_x(f_1' \dots f_{m'}') \leq (1 - \varepsilon_0)B_{j',m'}, \\ \log_x(f_1 \dots f_m e) \leq (1 - \varepsilon_0)(A_j - \varepsilon), & \log_x(f_1' \dots f_{m'}' e') \leq (1 - \varepsilon_0)(A_{j'} + \varepsilon), \\ ee' \text{ is } x^\delta\text{-smooth,} & \log_x(f_i), \log_x(f_i') \geq \delta \text{ for all } i \end{array} \right\}$$

We then say the moduli corresponding to region $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ are the set

$$Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0) = \bigcup_{j=1}^n \bigcup_{j'=1}^n \bigcup_{m=0}^{\lfloor \frac{1}{\delta} \rfloor} \bigcup_{m'=0}^{\lfloor \frac{1}{\delta} \rfloor} Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0).$$

Unlike in Polymath's proof, we will not always work directly with primes in arithmetic progressions. Instead we can replace the prime indicator function with a suitable minorant $\rho(n) \leq 1_{\mathbb{P}}(n)$, constructed via Harman's sieve, which satisfies an equidistribution statement of the type described below.

Definition 3 (Equidistribution). *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1 and let $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ be as given in Definition 2.*

We say that a function $f : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{C}$ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ provided that the following holds: For any $x > 1$, $\varepsilon_0 > 0$ and $A > 0$, and for any $a \in \mathbb{Z}$ with $(a, p) = 1$ for all primes $p \leq x$, we have

$$\sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0) \\ q \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(q)}} f(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x, 2x] \\ (n, q) = 1}} f(n; x) \right| \ll_{A, \varepsilon_0} \frac{x}{\log(x)^A}.$$

Next we write down the GPY sieve with a minorant of the prime indicator function. Recall here that $(h_1, \dots, h_k)$ is an admissible k -tuple if $h_1 < \dots < h_k$ are integers which avoid at least one residue class modulo p for each prime p .

Definition 4 (GPY sieve). *Let $(h_1, \dots, h_k)$ be an admissible k -tuple.*

Let $W = \prod_{p \leq \log \log x} p$ and choose $b \in \mathbb{N}$ with $(b + h_i, W) = 1$ for all $i \in \{1, \dots, k\}$.

Let $\rho : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{R}$ with $\rho(n; x) \leq 1_{\mathbb{P}}(n)$ for all large x and $n \in [x, 2x]$.

For given $L \in \mathbb{N}$, $c_1, \dots, c_L \in \mathbb{R}$ and smooth compactly supported functions $F_{l,i} : [0, \infty) \rightarrow \mathbb{R}$, we define the following sieve weights:

$$\nu(n) = \left(\sum_{l=1}^L c_l \lambda_{F_{l,1}}(n + h_1) \dots \lambda_{F_{l,k}}(n + h_k) \right)^2 \quad \text{with} \quad \lambda_F(n) = \sum_{d|n} \mu(d) F(\log_x(d)).$$

For the chosen weight function $\nu : \mathbb{N} \rightarrow [0, \infty)$, we then define the GPY sieve function

$$N(x; \rho, \nu) = \sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W)}} \nu(n) \left(\sum_{i=1}^k \rho(n + h_i; x) - 1 \right).$$

If $N(x; \rho, \nu) > 0$ then there must exist some $n_0 \in [x, 2x]$ for which $\sum_{i=1}^k \rho(n_0 + h_i; x) > 1$. But then $(n_0 + h_1, \dots, n_0 + h_k)$ must contain at least two primes. Thus $N(x; \rho, \nu) > 0$ for large x would imply that $H_1 \leq h_k - h_1$. The difficulty is thus to find a choice of $F_{l,i}$ (and thus a choice of ν) for which we can prove the inequality $N(x; \rho, \nu) > 0$. This can be reduced to an optimization problem involving integrals:

Definition 5 (Key integrals). *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1. Let F be a symmetric, square-integrable function supported on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. Then define*

$$\begin{aligned} I(F; \delta, \underline{A}, \underline{B}, \varepsilon) &= \int \dots \int_{T_k(\delta, \underline{A}, \underline{B}, \varepsilon)} F(t_1, \dots, t_k)^2 dt_1 \dots dt_k, \\ J(F; \delta, \underline{A}, \underline{B}, \varepsilon) &= \sum_{m=1}^n \sum_{m'=1}^n \int \dots \int_{\substack{t_1 + \dots + t_{k-1} \leq \max\{A_m - \varepsilon, A_{m'} - \varepsilon\} \\ t_1 + \dots + t_{k-1} + t_k \in [A_{m-1} + \varepsilon, A_m + \varepsilon] \\ t_1 + \dots + t_{k-1} + t'_k \in [A_{m'-1} + \varepsilon, A_{m'} + \varepsilon]}} F(t_1, \dots, t_{k-1}, t_k) F(t_1, \dots, t_{k-1}, t'_k) dt_1 \dots dt_{k-1} dt_k dt'_k, \\ K(F; \delta, \underline{A}, \underline{B}, \varepsilon) &= \sum_{m=1}^n \sum_{m'=1}^n \int \dots \int_{\substack{t_1 + \dots + t_{k-1} > \max\{A_m - \varepsilon, A_{m'} - \varepsilon\} \\ t_1 + \dots + t_{k-1} + t_k \in [A_{m-1} + \varepsilon, A_m + \varepsilon] \\ t_1 + \dots + t_{k-1} + t'_k \in [A_{m'-1} + \varepsilon, A_{m'} + \varepsilon]}} F(t_1, \dots, t_k)^2 dt_1 \dots dt_k. \end{aligned}$$

2.2. Main sieve statement. We are now ready to state the main result of this section:

Proposition 1. *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1. Suppose that function $\rho : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{R}$ and constants c_1 and c_2 have the following properties:*

- (1) *Prime minorant: $-c_2 \leq \rho(n; x) \leq 1_{\mathbb{P}}(n)$ for all large x and $n \in [x, 2x]$.*
- (2) *ρ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. (See Definition 3.)*
- (3) *If $\rho(n; x) \neq 0$, then all prime factors of n exceed x^β , for some $\beta > \max\{B_{j,1} : 1 \leq j \leq n\}$.*
- (4) *The sum of $\rho(n; x)$ over $[x, 2x]$ satisfies $\sum_{n \in [x, 2x]} \rho(n; x) = (1 - c_1 + o(1)) \frac{x}{\log(x)}$.*

Further let $I(F; \delta, \underline{A}, \underline{B}, \varepsilon)$, $J(F; \delta, \underline{A}, \underline{B}, \varepsilon)$ and $K(F; \delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 5. Suppose there exists a symmetric, square-integrable function F supported on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ with

$$(2.1) \quad \frac{k(1 - c_1)J(F; \delta, \underline{A}, \underline{B}, \varepsilon) - kc_2K(F; \delta, \underline{A}, \underline{B}, \varepsilon)}{I(F; \delta, \underline{A}, \underline{B}, \varepsilon)} > 1.$$

Then $H_1 = \liminf_{n \rightarrow \infty} (p_{n+1} - p_n) \leq H(k)$, where $H(k)$ is the diameter of the shortest admissible k -tuple.

2.3. Construction of sieve weights. We first show how a function F satisfying (2.1) is used to construct sieve weights $\nu(n)$. Our arguments follow the steps of Section 5 of Polymath [5], adapted to the new support $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$.

Lemma 1. *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as described in Definition 1. Let $\varepsilon_0 > 0$ and define region*

$$R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0) = \left\{ (t_1, \dots, t_k) \in [0, 1]^k : \begin{array}{l} \sum_{i=1}^k t_i < (1 - \varepsilon_0)(A_j + \varepsilon), \\ \sum_{i \in I} t_i < (1 - \varepsilon_0)B_{j,|I|} \text{ for } I = \{i : t_i > \delta\} \end{array} \right\}.$$

Suppose there exists a symmetric, square-integrable function F supported on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ with

$$\frac{k(1 - c_1)J(F; \delta, \underline{A}, \underline{B}, \varepsilon) - kc_2K(F; \delta, \underline{A}, \underline{B}, \varepsilon)}{I(F; \delta, \underline{A}, \underline{B}, \varepsilon)} > 1.$$

If $\varepsilon_0 > 0$ is chosen sufficiently small, we can then find integers $L_1, \dots, L_n$, constants $c_{j,l}$ and smooth compactly supported functions $f_{j,l,i} : [0, \infty) \rightarrow \mathbb{R}$ with the following two properties:

- (1) *For every $j \in \{1, \dots, n\}$ and $l \in \{1, \dots, L_j\}$, the support of product $f_{j,l,1}(t_1) \dots f_{j,l,k}(t_k)$ is contained in $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0)$.*
- (2) *Denoting by $\mathcal{L}(j, i)$ the set of $l \in \{1, \dots, L_j\}$ for which the support of product $\prod_{s \neq i} f_{j,l,s}(t_s)$ is contained in $\{(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k) : \sum_{s \neq i} t_s < (1 - \varepsilon_0)(A_j - \varepsilon)\}$, and denoting by $\mathcal{U}(j, i)$ its complement $\{1, \dots, L_j\} \setminus \mathcal{L}(j, i)$, the terms*

$$\begin{aligned} \mathcal{I} &= \sum_{j=1}^n \sum_{l=1}^{L_j} \sum_{j'=1}^n \sum_{l'=1}^{L_{j'}} c_{j,l} c_{j',l'} \prod_{s=1}^k \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s, \\ \mathcal{J}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j',i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \prod_{s \neq i} \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \\ &\quad + 2 \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j',i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \prod_{s \neq i} \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \\ \mathcal{K}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{U}(j',i)} c_{j,l} c_{j',l'} \prod_{s=1}^k \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \end{aligned}$$

satisfy the inequality

$$\frac{(1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i}{\mathcal{I}} > 1.$$

Our choice of sieve weights in the GPY method will then be as follows:

$$\nu(n) = \left(\sum_{j=1}^n \sum_{l=1}^{L_j} c_{j,l} \lambda_{f_{j,l,1}}(n + h_1) \dots \lambda_{f_{j,l,k}}(n + h_k) \right)^2$$

Proof. Let $F : [0, \infty)^k \rightarrow \mathbb{R}$ be a symmetric, square-integrable function, supported on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, which satisfies the inequality

$$(2.2) \quad \frac{k(1 - c_1)J(F; \delta, \underline{A}, \underline{B}, \varepsilon) - kc_2K(F; \delta, \underline{A}, \underline{B}, \varepsilon)}{I(F; \delta, \underline{A}, \underline{B}, \varepsilon)} > 1.$$

Let $S_j = \{(t_1, \dots, t_k) \in [0, 1]^k : \sum_{i=1}^k t_i \in [A_{j-1} + \varepsilon, A_j + \varepsilon) \text{ and } \sum_{i \in I} t_i \leq B_{j,|I|} \text{ for } I = \{i : t_i > \delta\}\}$ and let $1_{S_j}(n)$ be the indicator function of set S_j . Since $\{S_1, \dots, S_n\}$ is a partition of $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, we then have $F = \sum_{j=1}^n F \cdot 1_{S_j}$.

Next we choose small $\varepsilon_1, \varepsilon_2 > 0$ and define $F_{2,j} : [0, \infty)^k \rightarrow \mathbb{R}$ as follows:

$$F_{2,j}(t_1, \dots, t_k) = \begin{cases} F\left(\frac{t_1 - \varepsilon_2}{1 - \varepsilon_1}, \dots, \frac{t_k - \varepsilon_2}{1 - \varepsilon_1}\right) \cdot 1_{S_j}\left(\frac{t_1 - \varepsilon_2}{1 - \varepsilon_1}, \dots, \frac{t_k - \varepsilon_2}{1 - \varepsilon_1}\right) & \text{if } t_1, \dots, t_k \geq \varepsilon_2, \\ 0 & \text{otherwise.} \end{cases}$$

We also take $\varepsilon_0 = \frac{\varepsilon_1}{100}$ and define the integrals

$$\begin{aligned} I(F, G) &= \int_0^\infty \dots \int_0^\infty F(t_1, \dots, t_k) G(t_1, \dots, t_k) dt_1 \dots dt_k, \\ J_i(F, G; B) &= \int_0^\infty \dots \int_0^\infty F(t_1, \dots, t_{i-1}, t_i, t_{i+1}, \dots, t_k) G(t_1, \dots, t_{i-1}, t'_i, t_{i+1}, \dots, t_k) dt'_i dt_1 \dots dt_k, \\ &\quad \sum_{s \neq i} t_s < B \\ K_i(F, G; B) &= \int_0^\infty \dots \int_0^\infty F(t_1, \dots, t_k) G(t_1, \dots, t_k) dt_1 \dots dt_k, \\ &\quad \sum_{s \neq i} t_s > B \end{aligned}$$

Since F is square-integrable and symmetric, and satisfies inequality (2.2), we can choose a very small $\varepsilon_1 > 0$ and an even smaller $\varepsilon_2 > 0$ so that our shifted and scaled functions $F_{2,j}$ still satisfy

$$(2.3) \quad \frac{\sum_{i=1}^k \sum_{j=1}^n \sum_{j'=1}^n ((1 - c_1) J_i(F_{2,j}, F_{2,j'}; B^*(j, j')) - c_2 K_i(F_{2,j}, F_{2,j'}; B^*(j, j')))}{\sum_{j=1}^n \sum_{j'=1}^n I(F_{2,j}, F_{2,j'})} > 1$$

where $B^*(j, j') = (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}$.

Furthermore, recall the assumption $B_{j,m} \leq B_{j,m+1} \leq B_{j,m} + \delta$. Hence $\sum_{i \in I} t_i \leq B_{j,|I|}$ for $I = \{i : t_i > \delta\}$ implies $\sum_{i \in I'} t_i \leq B_{j,|I'|}$ for any $I' \subseteq I$. In particular, if $(t_1, \dots, t_k)$ is in the support of $F_{2,j}$, and $L = \{i : t_i > \delta\}$, then for any $i \in L$, also $\frac{t_i - \varepsilon_2}{1 - \varepsilon_1} > \delta$ and so $\sum_{i \in L} (\frac{t_i - \varepsilon_2}{1 - \varepsilon_1}) \leq B_{j,|L|}$. Hence, for ε_2 much smaller than ε_1 , we see that the support of $F_{2,j}$ is contained inside $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, 2\varepsilon_0)$, and it actually stays away from all boundary faces of $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, 2\varepsilon_0)$. Thus we can convolve each $F_{2,j}$ with a smooth approximation to the identity, supported close to the origin, to obtain functions $F_{3,j} : [0, \infty)^k \rightarrow \mathbb{R}$ which are smooth, supported on $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, 2\varepsilon_0)$ and satisfy the inequality

$$(2.4) \quad \frac{\sum_{i=1}^k \sum_{j=1}^n \sum_{j'=1}^n ((1 - c_1) J_i(F_{3,j}, F_{3,j'}; B^*(j, j')) - c_2 K_i(F_{3,j}, F_{3,j'}; B^*(j, j')))}{\sum_{j=1}^n \sum_{j'=1}^n I(F_{3,j}, F_{3,j'})} > 1.$$

Following the arguments of Polymath [5], we next define $f_{3,j} : [0, \infty)^k \rightarrow \mathbb{R}$ by setting

$$f_{3,j}(t_1, \dots, t_k) = \int_{t_1}^\infty \dots \int_{t_k}^\infty F_{3,j}(s_1, \dots, s_k) ds_1 \dots ds_k.$$

Yet again, the condition $B_{j,m} \leq B_{j,m+1} \leq B_{j,m} + \delta$ ensures that the support of $f_{3,j}$ stays contained in $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, 2\varepsilon_0)$. Furthermore,

$$\begin{aligned} I(F_{3,j}, F_{3,j'}) &= \int_0^\infty \dots \int_0^\infty \frac{\partial^k}{\partial t_1 \dots \partial t_k} f_{3,j}(t_1, \dots, t_k) \frac{\partial^k}{\partial t_1 \dots \partial t_k} f_{3,j'}(t_1, \dots, t_k) dt_1 \dots dt_k, \\ J_i(F_{3,j}, F_{3,j'}; B) &= \int_0^\infty \dots \int_0^\infty \frac{\partial^{k-1}}{\prod_{s \neq i} \partial t_s} f_{3,j}(t_1, \dots, t_{i-1}, 0, t_{i+1}, \dots, t_k) \frac{\partial^{k-1}}{\prod_{s \neq i} \partial t_s} f_{3,j'}(t_1, \dots, t_{i-1}, 0, t_{i+1}, \dots, t_k) \prod_{s \neq i} dt_s, \\ K_i(F_{3,j}, F_{3,j'}; B) &= \int_0^\infty \dots \int_0^\infty \frac{\partial^k}{\partial t_1 \dots \partial t_k} f_{3,j}(t_1, \dots, t_k) \frac{\partial^k}{\partial t_1 \dots \partial t_k} f_{3,j'}(t_1, \dots, t_k) dt_1 \dots dt_k. \\ &\quad \sum_{s \neq i} t_s > B \end{aligned}$$

We now refer to $I(F_{3,j}, F_{3,j'})$ by $I^*(f_{3,j}, f_{3,j'})$, we refer to $J_i(F_{3,j}, F_{3,j'}; B)$ by $J_i^*(f_{3,j}, f_{3,j'}; B)$ and we refer to $K_i(F_{3,j}, F_{3,j'}; B)$ by $K_i^*(f_{3,j}, f_{3,j'}; B)$. Hence

$$(2.5) \quad \frac{\sum_{i=1}^k \sum_{j=1}^n \sum_{j'=1}^n ((1 - c_1) J_i^*(f_{3,j}, f_{3,j'}; B^*(j, j')) - c_2 K_i^*(f_{3,j}, f_{3,j'}; B^*(j, j')))}{\sum_{j=1}^n \sum_{j'=1}^n I^*(f_{3,j}, f_{3,j'})} > 1$$

where $B^*(j, j') = (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}$.

Next we choose another very small constant $\varepsilon_3 > 0$ and (like in Section 5.3 of [5]) apply the Stone-Weierstrass theorem to each function $f_{3,j}$, expressing it as the smooth limit of linear combinations of the form $\sum c_l f_{4,j,l,1}(t_1) \dots f_{4,j,l,k}(t_k)$, where each $f_{4,j,l,i}(t_i)$ is a smooth function supported on an interval of length at most ε_3 . Overall, we can find integers $L_1, \dots, L_n$, constants $c_{j,l}$ and smooth functions $f_{j,l,i} : [0, \infty) \rightarrow \mathbb{R}$ with the following properties:

- (i) Each $f_{j,l,i}(t)$ is smooth and supported on an interval of length at most ε_3 .
- (ii) For every $j \in \{1, \dots, n\}$ and $l \in \{1, \dots, L_j\}$, the support of product $f_{j,l,1}(t_1) \dots f_{j,l,k}(t_k)$ is contained in $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0)$.
- (iii) For each $j \in \{1, \dots, n\}$, sum

$$h_j(t_1, \dots, t_k) = \sum_{l=1}^{L_j} c_{j,l} f_{j,l,1}(t_1) \dots f_{j,l,k}(t_k)$$

approximates function $f_{3,j}(t_1, \dots, t_k)$ so well that the following is true:

$$(2.6) \quad \frac{\sum_{i=1}^k \sum_{j=1}^n \sum_{j'=1}^n ((1 - c_1) J_i^*(h_j, h_{j'}; B^*(j, j')) - c_2 K_i^*(h_j, h_{j'}; B^*(j, j')))}{\sum_{j=1}^n \sum_{j'=1}^n I^*(h_j, h_{j'})} > 1.$$

Now we take a closer look at I^* , J_i^* and K_i^* . The expression $\sum_{j=1}^n \sum_{j'=1}^n I^*(h_j, h_{j'})$ rearranges to

$$\sum_{j=1}^n \sum_{l=1}^{L_j} \sum_{j'=1}^n \sum_{l'=1}^{L_{j'}} c_{j,l} c_{j',l'} \int_0^\infty \dots \int_0^\infty f'_{j,l,1}(t_1) \dots f'_{j,l,k}(t_k) f'_{j',l',1}(t_1) \dots f'_{j',l',k}(t_k) dt_1 \dots dt_k.$$

But this is simply $\mathcal{I}$. On the other hand, consider $\sum_{j=1}^n \sum_{j'=1}^n J_i^*(h_j, h_{j'}; (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\})$ for some $i \in \{1, \dots, k\}$. It equals the following expression:

$$\mathcal{J}_i^* = \sum_{j=1}^n \sum_{l=1}^{L_j} \sum_{j'=1}^n \sum_{l'=1}^{L_{j'}} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \int_0^\infty \dots \int_0^\infty \prod_{\substack{s \neq i \\ \sum_{s \neq i} t_s < (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}}} f'_{j,l,s}(t_s) \prod_{\substack{s \neq i}} f'_{j',l',s}(t_s) \prod_{s \neq i} dt_s.$$

We now partition each $\{1, \dots, L_j\}$ as follows: $\mathcal{L}(j, i)$ is the set of $l \in \{1, \dots, L_j\}$ for which the support of product $\prod_{s \neq i} f_{j,l,s}(t_s)$ is contained in $\{(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k) : \sum_{s \neq i} t_i < (1 - \varepsilon_0)(A_j - \varepsilon)\}$. The set $\mathcal{U}(j, i)$ is the complement $\{1, \dots, L_j\} \setminus \mathcal{L}(j, i)$.

The union of the supports of functions $\prod_{s \neq i} f_{j,l,s}(t_s) \prod_{s \neq i} f_{j',l',s}(t_s)$ with $l \in \mathcal{U}(j, i)$ and $l' \in \mathcal{U}(j', i)$ only overlaps with $\{(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k) : \sum_{s \neq i} t_i < (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}\}$ on a set of measure $O(\varepsilon_3)$. Hence for any given $\varepsilon_4 > 0$, we can choose the value of ε_3 sufficiently small that the contribution of pairs $(l, l') \in \mathcal{U}(j, i) \times \mathcal{U}(j', i)$ to $\mathcal{J}_i^*$ is bounded by ε_4 and

$$\begin{aligned} \mathcal{J}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{L}(j, i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j', i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \int_0^\infty \dots \int_0^\infty \prod_{\substack{s \neq i}} f'_{j,l,s}(t_s) \prod_{\substack{s \neq i}} f'_{j',l',s}(t_s) \prod_{s \neq i} dt_s \\ &+ 2 \sum_{j=1}^n \sum_{l \in \mathcal{U}(j, i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j', i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \int_0^\infty \dots \int_0^\infty \prod_{\substack{s \neq i}} f'_{j,l,s}(t_s) \prod_{\substack{s \neq i}} f'_{j',l',s}(t_s) \prod_{s \neq i} dt_s \\ &> \sum_{j=1}^n \sum_{j'=1}^n J_i^*(h_j, h_{j'}; (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}) - \varepsilon_4. \end{aligned}$$

Finally, pairs (l, l') with $l \in \mathcal{L}(j, i)$ or $l' \in \mathcal{L}(j', i)$ do not contribute at all to $\sum_{j=1}^n \sum_{j'=1}^n K_i^*(h_j, h_{j'}; B^*(j, j'))$, as their support does not overlap with the integration region. Thus the sum of K_i^* equals

$$\mathcal{K}_i^* = \sum_{j=1}^n \sum_{l \in \mathcal{U}(j, i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{U}(j', i)} c_{j,l} c_{j',l'} \int_0^\infty \dots \int_0^\infty \prod_{s=1}^k f'_{j,l,s}(t_s) \prod_{s=1}^k f'_{j',l',s}(t_s) dt_1 \dots dt_k.$$

Since the support of functions $\prod_{s \neq i} f_{j,l,i}(t_i) \prod_{s \neq i} f_{j',l',i}(t_i)$ with $l \in \mathcal{U}(j, i)$ and $l' \in \mathcal{U}(j', i)$ only overlaps with $\{(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k) : \sum_{s \neq i} t_i < (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}\}$ on a set of measure $O(\varepsilon_3)$, we can drop the condition $\sum_{s \neq i} t_s > (1 - \varepsilon_0) \max\{A_j - \varepsilon, A_{j'} - \varepsilon\}$ with a loss of at most ε_4 , provided that

ε_3 was chosen small enough. Overall we then have

$$\begin{aligned} \mathcal{K}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{U}(j',i)} c_{j,l} c_{j',l'} \int_0^\infty \cdots \int_0^\infty \prod_{s=1}^k f'_{j,l,s}(t_s) \prod_{s=1}^k f'_{j',l',s}(t_s) dt_1 \dots dt_k \\ &< \sum_{j=1}^n \sum_{j'=1}^n K_i^*(h_j, h_{j'}; B^*(j, j')) + \varepsilon_4. \end{aligned}$$

Looking back at inequality (2.6) we therefore have

$$\frac{(1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i + ((1 - c_1)k + c_2 k) \varepsilon_4}{\mathcal{I}} > 1.$$

Hence, taking ε_4 sufficiently small (and the corresponding ε_3 even smaller), we arrive at the desired inequality $\frac{(1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i}{\mathcal{I}} > 1$. $\square$

2.4. Asymptotics. Having chosen sieve weights $\nu(n)$, we now need to prove asymptotics for sums $\sum_{x \leq n \leq 2x} \nu(n) \sum_{i=1}^k \rho(n + h_i; x)$. We first look at $\sum_{x \leq n \leq 2x} \rho(n + h_i; x) \prod_{s=1}^k \lambda_{F_s}(n + h_s) \lambda_{G_s}(n + h_s)$. Here we can follow the arguments of Section 4.2 of Polymath [5], with small amendments due to use of Harman's sieve (like in Baker and Irving's work [1]) and due to the new support.

Lemma 2. *Suppose that function $\rho : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{R}$ has the following properties:*

- (1) *Prime minorant: $-c_2 \leq \rho(n; x) \leq 1_{\mathbb{P}}(n)$ for all large x and $n \in [x, 2x]$.*
- (2) *ρ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. (See Definition 3.)*
- (3) *If $\rho(n; x) \neq 0$, then all prime factors of n exceed x^β , for some $\beta > \max\{B_{j,1} : 1 \leq j \leq n\}$.*
- (4) *The sum of $\rho(n; x)$ over $[x, 2x]$ satisfies $\sum_{n \in [x, 2x]} \rho(n; x) = (1 - c_1 + o(1)) \frac{x}{\log(x)}$.*

Let $i_0 \in \{1, \dots, k\}$. Consider smooth compactly supported functions $F_i : [0, \infty) \rightarrow \mathbb{R}$ and $G_i : [0, \infty) \rightarrow \mathbb{R}$. Let $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0)$ be as defined in Lemma 1. Suppose that the support of product $F_1(t_1) \dots F_k(t_k)$ is contained in the set $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0)$ and the support of product $G_1(t_1) \dots G_k(t_k)$ is contained in $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j', \varepsilon_0)$. Finally, suppose that the support of product $\prod_{i \neq i_0} F_i(t_i)$ is contained in the set $\{(t_1, \dots, t_{i_0-1}, t_{i_0+1}, \dots, t_k) : \sum_{i \neq i_0} t_i < (1 - \varepsilon_0)(A_j - \varepsilon)\}$.

Let $W = \prod_{p \leq \log \log \log x} p$ and $b \in \mathbb{N}$ with $(b + h_i, W) = 1$ for all $i \in \{1, \dots, k\}$. Then

$$\begin{aligned} &\sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W)}} \rho(n + h_{i_0}; x) \prod_{i=1}^k \lambda_{F_i}(n + h_i) \lambda_{G_i}(n + h_i) \\ &= \left((1 - c_1) F_k(0) G_k(0) \prod_{i \neq i_0} \int_0^\infty F'_i(t_i) G'_i(t_i) dt_i + o(1) \right) \left(\frac{x W^{k-1}}{\phi(W)^k \log(x)^k} \right). \end{aligned}$$

Proof. Without loss of generality we assume that $i_0 = k$. Recall that $\lambda_F(n) = \sum_{d|n} \mu(d) F(\log_x(d))$. We call the sum we are interested in S . Expanding, we have

$$\begin{aligned} S &:= \sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W)}} \rho(n + h_k; x) \prod_{i=1}^k \lambda_{F_i}(n + h_i) \lambda_{G_i}(n + h_i) \\ &= \sum_{d_1, \dots, d_k, d'_1, \dots, d'_k} \prod_{i=1}^k \mu(d_i) \mu(d'_i) F_i(\log_x(d_i)) G_i(\log_x(d'_i)) \sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W) \\ n \equiv -h_i[d_i, d'_i] \forall i}} \rho(n + h_k; x). \end{aligned}$$

If $\rho(n + h_k; x) \neq 0$, then any $d \mid n + h_k$ with $d \neq 1$ satisfies $\log_x(d) > B_{j,1}$. But by construction we have $B_{j,m} \leq B_{j,m+1} \leq B_{j,m} + \delta$, and so $\log_x(d) > B_{j,1}$ implies $\log_x(d) + (m-1)\delta > B_{j,m}$ for all m . Thus any tuple $(\log_x(d_1), \dots, \log_x(d_k))$ with $(d_k \mid n + h_k \text{ and } d_k \neq 1)$ cannot be in the support of product $F_1(t_1) \dots F_k(t_k)$. Only $d_k = 1$ contributes to S . Similarly, only $d'_k = 1$ contributes to S and so we have

$$S = F_k(0) G_k(0) \sum_{d_1, \dots, d_k, d'_1, \dots, d'_k} \prod_{i=1}^{k-1} \mu(d_i) \mu(d'_i) F_i(\log_x(d_i)) G_i(\log_x(d'_i)) \sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W) \\ n \equiv -h_i[d_i, d'_i] \forall i \neq k}} \rho(n + h_k; x).$$

Now we use the W -trick as in Polymath [5]: b was chosen to satisfy $(b+h_i, W) = 1$ for all i . Assuming that x is sufficiently large to ensure $|h_i - h_{i'}| < \log \log \log x$ for $i \neq i'$, the system of congruences given above thus can only have solutions if the lcms $[d_i, d'_i]$ are coprime to each other and to $W = \prod_{p \leq \log \log \log x} p$. The conditions $(n \equiv b(W) \text{ and } n \equiv -h_i[d_i, d'_i] \text{ for } i \neq k)$ can thus be replaced by a single congruence $n + h_k \equiv a_{W, d_1, \dots, d'_{k-1}} \pmod{q_{W, d_1, \dots, d'_{k-1}}}$. Here $a_{W, d_1, \dots, d'_{k-1}}$ can be chosen to be coprime to all primes $p \leq x$, and $q_{W, d_1, \dots, d'_{k-1}}$ is defined to equal

$$q_{W, d_1, \dots, d'_{k-1}} = W[d_1, d'_1] \dots [d_{k-1}, d'_{k-1}].$$

Hence we split S into a main term S_M and an error term S_E as follows:

$$S_M = F_k(0)G_k(0) \sum_{d_1, \dots, d_k, d'_1, \dots, d'_k} \frac{\prod_{i=1}^{k-1} \mu(d_i)\mu(d'_i)F_i(\log_x(d_i))G_i(\log_x(d'_i))}{\phi(W[d_1, d'_1] \dots [d_{k-1}, d'_{k-1}])} \sum_{x+h_k \leq m \leq 2x+h_k} \rho(m; x),$$

$$S_E = |F_k(0)G_k(0)| \sum_{\substack{d_1, \dots, d_k, d'_1, \dots, d'_k \\ q_{W, d_1, \dots, d'_{k-1}} \text{ squarefree}}} \prod_{i=1}^{k-1} |F_i(\log_x(d_i))G_i(\log_x(d'_i))| \Delta(\rho; a_{W, d_1, \dots, d'_{k-1}}(q_{W, d_1, \dots, d'_{k-1}})),$$

where $\Delta(\rho; a(q)) = \left| \sum_{\substack{x+h_k \leq m \leq 2x+h_k \\ m \equiv a(q)}} \rho(m; x) - \frac{1}{\phi(q)} \sum_{\substack{x+h_k \leq m \leq 2x+h_k \\ (m, q)=1}} \rho(m; x) \right|$.

To deal with the main term S_M , we recall assumption $\sum_{n \in [x, 2x]} \rho(n; x) = (1 - c_1 + o(1)) \frac{x}{\log(x)}$, and we apply Lemma 4.1 of Polymath [5] with $[d_i, d'_i]$ replaced by $\phi([d_i, d'_i])$ (see also the discussion on page 19 of [5]). Overall this gives the asymptotic

$$S_M = \left(F_k(0)G_k(0) \left(\prod_{i=1}^{k-1} \int_0^\infty F'_i(t_i)G'_i(t_i)dt_i \right) (1 - c_1) + o(1) \right) \left(\frac{xW^{k-1}}{\phi(W)^k \log(x)^k} \right).$$

To deal with the error term S_E , we first recall that the support of product $G_1(t_1) \dots G_k(t_k)$ is contained in $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j', \varepsilon_0)$ and that the support of product $\prod_{i \neq i_0} F_i(t_i)$ is contained in the set $\{(t_1, \dots, t_{i_0-1}, t_{i_0+1}, \dots, t_k) : \sum_{i \neq i_0} t_i < (1 - \varepsilon_0)(A_j - \varepsilon)\}$. Consider modulus $q = W[d_1, d'_1] \dots [d_{k-1}, d'_{k-1}]$, and assume the modulus contributes to S_E . Relabel those d_i which satisfy $d_i \geq x^\delta$ as f_i and combine the remaining d_i and W to a product called e . Relabel $d'_i/(d_i, d'_i)$ which satisfy $d'_i/(d_i, d'_i) \geq x^\delta$ as f'_i and combine the remaining $d'_i/(d_i, d'_i)$ to a product called e' . Then we have $q = ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i$, where e and e' are x^δ -smooth and $\log_x(f_1 \dots f_m e) \leq (1 - \frac{\varepsilon_0}{2})(A_j - \varepsilon)$ and $\log_x(f'_1 \dots f'_{m'} e') \leq (1 - \frac{\varepsilon_0}{2})(A_{j'} + \varepsilon)$. Using once more that $B_{j, m} \leq B_{j, m+1} \leq B_{j, m} + \delta$, we also have $\log_x(f_1 \dots f_m) \leq (1 - \varepsilon_0)B_{j, m}$ and $\log_x(f'_1 \dots f'_{m'}) \leq (1 - \varepsilon_0)B_{j', m'}$. In short, comparing with Definition 2, we see that any modulus $q = W[d_1, d'_1] \dots [d_{k-1}, d'_{k-1}]$ which contributes to S_E is contained in the set $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2})$.

We then recall our assumption (2) that ρ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$: If $a \in \mathbb{Z}$ with $(a, p) = 1$ for all $p \leq x$, and if $C > 0$, we have

$$(2.7) \quad \sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2}) \\ q \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(q)}} \rho(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x, 2x] \\ (n, q)=1}} \rho(n; x) \right| \ll_{C, \varepsilon_0} \frac{x}{\log(x)^C}.$$

However, in the statement above, the congruence class a is fixed, whereas in S_E we have that $a_{W, d_1, \dots, d'_{k-1}}$ depends on d_i and d'_i . To get around this issue, we use a trick similar to that on page 22 of [5], and first define the following set:

$$\mathcal{A} = \left\{ a \in \left\{ 1, \dots, \prod_{p \leq x} p \right\} : a \equiv b + h_k \pmod{W}, \forall p \in (\log \log \log x, x] \exists i \in \{1, \dots, k-1\} : a \equiv h_k - h_i \pmod{p} \right\}.$$

First we think about a fixed choice of $d_1, \dots, d'_{k-1}$. The set $\mathcal{A}$ is constructed so that for any $q = q_{W, d_1, \dots, d'_{k-1}} = W[d_1, d'_1] \dots [d_{k-1}, d'_{k-1}]$, the corresponding $a_{W, d_1, \dots, d'_{k-1}}(q)$ is contained in $\mathcal{A}$. More specifically, by applying the Chinese Remainder Theorem, we see that when $\tilde{a}$ ranges over $\mathcal{A}$, then $\tilde{a} \equiv a_{W, d_1, \dots, d'_{k-1}}(q)$ for $|\mathcal{A}|/(k-1)^{\omega(q/W)}$ choices of $\tilde{a}$, where $\omega(q/W)$ is the number of distinct prime factors of q/W . But since q/W is squarefree, we have $(k-1)^{\omega(q/W)} \ll \tau(q)^{O(1)}$. Hence if we sum over all $\tilde{a} \in \mathcal{A}$ and multiply with $\frac{\tau(q)^{O(1)}}{|\mathcal{A}|}$, then $a_{W, d_1, \dots, d'_{k-1}}$ is counted at least once. Furthermore, any

$q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2})$ has $q = q_{W, d_1, \dots, d'_{k-1}}$ for at most $\tau(q)^{O(1)}$ choices of $d_1, \dots, d'_{k-1}$. Therefore we obtain the following bound on S_E :

$$S_E \ll \sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2}) \\ q \text{ squarefree}}} \frac{\tau(q)^{O(1)}}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \left| \sum_{\substack{n \in [x+h_k, 2x+h_k] \\ n \equiv a(q)}} \rho(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x+h_k, 2x+h_k] \\ (n, q)=1}} \rho(n; x) \right|.$$

To finish the proof we simply apply the Cauchy-Schwarz inequality and equidistribution property (2.7). Choosing constant C sufficient large, we have:

$$\begin{aligned} S_E &\ll \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \left(\sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2}) \\ q \text{ squarefree}}} \tau(q)^{O(1)} \Delta(\rho, a(q)) \right)^{1/2} \left(\sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \frac{\varepsilon_0}{2}) \\ q \text{ squarefree}}} \Delta(\rho, a(q)) \right)^{1/2} \\ &\ll_C \frac{1}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} (x \log(x)^{O(1)})^{1/2} (x \log(x)^{-C})^{1/2} \ll_C \frac{x}{\log(x)^{C/2}}. \end{aligned}$$

Since $W \ll (\log \log x)^{O(1)}$, summing up the asymptotic for S_M and the bound on S_E for a large C completes the proof. $\square$

2.5. Proof of Proposition 1. Proposition 1 can now be obtained as a corollary of Lemma 1, Lemma 2, as well as Theorem 3.6 of [5]. We need to be slightly careful when mixing Harman's sieve ($\rho(n; x)$) and the epsilon enlargement trick.

Proof of Proposition 1. Let $\varepsilon_0 > 0$ be very small. By Lemma 1, we can then find integers $L_1, \dots, L_n$, constants $c_{j,l}$ and smooth compactly supported functions $f_{j,l,i} : [0, \infty) \rightarrow \mathbb{R}$, with $f_{j,l,1}(t_1) \dots f_{j,l,k}(t_k)$ supported on $R_k^+(\delta, \underline{A}, \underline{B}, \varepsilon, j, \varepsilon_0)$, so that the following is true: Denoting by $\mathcal{L}(j, i)$ the set of $l \in \{1, \dots, L_j\}$ for which the support of product $\prod_{s \neq i} f_{j,l,s}(t_s)$ is contained in $\{(t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k) : \sum_{s \neq i} t_i < (1 - \varepsilon_0)(A_j - \varepsilon)\}$, and denoting by $\mathcal{U}(j, i)$ its complement $\{1, \dots, L_j\} \setminus \mathcal{L}(j, i)$, the terms

$$\begin{aligned} \mathcal{I} &= \sum_{j=1}^n \sum_{l=1}^{L_j} \sum_{j'=1}^n \sum_{l'=1}^{L_{j'}} c_{j,l} c_{j',l'} \prod_{s=1}^k \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s, \\ \mathcal{J}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j',i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \prod_{s \neq i} \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \\ &\quad + 2 \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{L}(j',i)} c_{j,l} c_{j',l'} f_{j,l,i}(0) f_{j',l',i}(0) \prod_{s \neq i} \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \\ \mathcal{K}_i &= \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} \sum_{j'=1}^n \sum_{l' \in \mathcal{U}(j',i)} c_{j,l} c_{j',l'} \prod_{s=1}^k \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s \end{aligned}$$

satisfy the inequality

$$\frac{(1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i}{\mathcal{I}} > 1.$$

We now choose the following sieve weights:

$$\nu(n) = \left(\sum_{j=1}^n \sum_{l=1}^{L_j} c_{j,l} \lambda_{f_{j,l,1}}(n + h_1) \dots \lambda_{f_{j,l,k}}(n + h_k) \right)^2.$$

Next we consider the ratio

$$\tilde{N}(x; \rho, \nu) = \frac{\sum_{i=1}^k \sum_{x \leq n \leq 2x, n \equiv b(W)} \nu(n) \rho(n + h_i; x)}{\sum_{x \leq n \leq 2x, n \equiv b(W)} \nu(n)}.$$

On the one hand, Theorem 3.6 of [5] tells us that

$$\sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W)}} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \lambda_{f_{j',l',s}}(n + h_s) = \left(\prod_{s=1}^k \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s + o(1) \right) \left(\frac{x W^{k-1}}{\phi(W)^k \log(x)^k} \right).$$

Expanding the square in $\nu(n)$, this gives

$$(2.8) \quad \sum_{x \leq n \leq 2x, n \equiv b(W)} \nu(n) = (\mathcal{I} + o(1)) \left(\frac{xW^{k-1}}{\phi(W)^k \log(x)^k} \right).$$

On the other hand, a lower bound for the enumerator is obtained as follows: $\rho(n + h_i; x) + c_2 \geq 0$ and

$$\begin{aligned} \nu(n)\rho(n + h_i; x) &\geq \left(\sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right)^2 (\rho(n + h_i; x) + c_2) \\ &\quad + 2 \left(\sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right) \left(\sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right) (\rho(n + h_i; x) + c_2) \\ &\quad - c_2 \left(\sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) + \sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right)^2 \\ &= \left(\sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right)^2 \rho(n + h_i; x) \\ &\quad + 2 \left(\sum_{j=1}^n \sum_{l \in \mathcal{L}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right) \left(\sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right) \rho(n + h_i; x) \\ &\quad - c_2 \left(\sum_{j=1}^n \sum_{l \in \mathcal{U}(j,i)} c_{j,l} \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \right)^2. \end{aligned}$$

But Lemma 2 gives us asymptotic

$$\begin{aligned} &\sum_{\substack{x \leq n \leq 2x \\ n \equiv b(W)}} \rho(n + h_i; x) \prod_{s=1}^k \lambda_{f_{j,l,s}}(n + h_s) \lambda_{f_{j',l',s}}(n + h_s) \\ &= \left((1 - c_1) f_{j,l,k}(0) f_{j',l',k}(0) \prod_{s \neq i} \int_0^\infty f'_{j,l,s}(t_s) f'_{j',l',s}(t_s) dt_s + o(1) \right) \left(\frac{xW^{k-1}}{\phi(W)^k \log(x)^k} \right). \end{aligned}$$

Summing the lower bound for $\nu(n)\rho(n + h_i; x)$ over $[x, 2x]$ and substituting the asymptotic from Lemma 2 for terms involving $\rho(n + h_i; x)$, and substituting the asymptotic from Theorem 3.6 of [5] for terms involving c_2 , we then get

$$(2.9) \quad \sum_{i=1}^k \sum_{x \leq n \leq 2x, n \equiv b(W)} \nu(n)\rho(n + h_i; x) = \left((1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i + o(1) \right) \left(\frac{xW^{k-1}}{\phi(W)^k \log(x)^k} \right).$$

Combining asymptotics (2.8) and (2.9), we finally get

$$\tilde{N}(x; \rho, \nu) = \frac{(1 - c_1) \sum_{i=1}^k \mathcal{J}_i - c_2 \sum_{i=1}^k \mathcal{K}_i + o(1)}{\mathcal{I} + o(1)} > 1 + o(1).$$

Hence, for sufficiently large x , we have the desired inequality $N(x; \rho, \nu) > 0$. $\square$

3. EQUIDISTRIBUTION ESTIMATES

In this section we derive equidistribution estimates for various convolutions of coefficient sequences. These will be needed to show that some minorant for the prime indicator function (denoted by $\rho(n; x)$) has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, provided that the entries of $\underline{B}$ are chosen small enough: If the entries of $\underline{B}$ are small, then the corresponding moduli, given by set $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$, are either smaller than $x^{1/2}$ or have a large x^δ -smooth factor. In the first case we can use the Bombieri-Vinogradov theorem, so we need to focus on moduli with a large x^δ -smooth factor.

Here the initial idea is to use the Type I/II/III estimates for i -tuply x^δ -densely divisible integers from Polymath's paper [4], the equidistribution estimates for x^δ -smooth moduli from the author's paper [6], and well as the Type I estimate of Baker and Irving [1]. However, there is a problem: a big x^δ -smooth factor is not sufficient to ensure that a number is i -tuply x^δ -densely divisible. Thus we need to prove

amended versions of the equidistribution estimates listed above. Fortunately we only need to make small changes in the relevant proofs.

3.1. Important definitions. Since this section discusses coefficient sequences, we first define these.

Definition 6 (Coefficient sequences). *A coefficient sequence is a function $\alpha : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{R}$ which satisfies $|\alpha(n; x)| \ll \tau(n)^{O(1)} \log(x)^{O(1)}$.*

- (i) α is said to be located at scale $N(x)$ if there are some constants $1 \ll c \ll C \ll 1$ such that $\alpha(\cdot; x)$ is supported on $[cN(x), CN(x)]$ for every $x > 1$.
- (ii) If α is located at scale $N(x)$, it is said to have the Siegel-Walfisz property if

$$\left| \sum_{\substack{n=a(q) \\ (n,r)=1}} \alpha(n; x) - \frac{1}{\phi(q)} \sum_{(n,qr)=1} \alpha(n; x) \right| \ll_A \tau(qr)^{O(1)} \frac{N(x)}{\log(x)^A}$$

for any $x > 1$, any $q, r \geq 1$, any $A > 1$, and any residue class $a \bmod q$ with $(a, q) = 1$.

- (iii) α is said to be smooth at scale $N(x)$ if there are some constants $1 \ll c \ll C \ll 1$ so that for every $x > 1$, there exists a smooth function $\psi : \mathbb{R} \rightarrow \mathbb{C}$ supported on $[c, C]$, with $|\psi^{(j)}(t)| \ll_j \log(x)^{O_j(1)}$ for all $t \in \mathbb{R}$, such that $\alpha(n; x) = \psi\left(\frac{n}{N}\right)$.

Further, as usually, convolutions of sequences are defined as follows: If $\alpha : \mathbb{N} \rightarrow \mathbb{C}$ and $\beta : \mathbb{N} \rightarrow \mathbb{C}$, then

$$(\alpha \star \beta)(n) = \sum_{d|n} \alpha(d) \beta\left(\frac{n}{d}\right).$$

We will now state various equidistribution estimates for Dirichlet convolutions $\alpha \star \beta$. Generalizing our earlier definition, we will use the following notion of equidistribution:

Definition 7 (Equidistribution). *Consider $f : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{C}$. Let $D(x; \omega, \gamma, \delta; \varepsilon)$ be a set of integers dependent on parameters ω, γ and δ , and dependent on a small parameter $\varepsilon > 0$.*

For a given set of (ω, γ, δ) , we say f has equidistribution properties for moduli in $D(x; \omega, \gamma, \delta; \varepsilon)$, provided the following holds: For any sufficiently small $\varepsilon > 0$ and for all $x > 1$ and $A > 0$, and for all $a \in \mathbb{Z}$ with $(a, p) = 1$ for $p \leq x$,

$$(3.1) \quad \sum_{\substack{d \in D(x; \omega, \gamma, \delta; \varepsilon) \\ d \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(d)}} f(n; x) - \frac{1}{\phi(d)} \sum_{\substack{n \in [x, 2x] \\ (n, d)=1}} f(n; x) \right| \ll_{A, \varepsilon} \frac{x}{\log(x)^A}.$$

Note that the implied constant is only allowed to depend on ω, γ and δ insofar that they determine which ε are sufficiently small for (3.1) to hold.

3.2. Equidistribution ingredients. To choose a suitable $\rho(n; x)$ and $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, we will use the following equidistribution estimates. Proofs follow in the next subsection.

Lemma 3 (Polymath Type II). *Suppose $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M and β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$ and $N(x) \leq x^{1/2}$. Suppose also that β has the Siegel-Walfisz property. Write $N(x) = x^\gamma$. Define the set of moduli*

$$D_{IIa}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \text{There exists a divisor } r \mid d \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \right\}.$$

Consider triples (ω, γ, δ) which satisfy the following inequalities:

$$\begin{aligned} 24\omega + 7\delta - 5\gamma &< -2, \\ 8\omega + 3\delta - \gamma &< 0. \end{aligned}$$

For such triples (ω, γ, δ) , convolution f has equidistribution properties for moduli in $D_{IIa}(x; \omega, \gamma, \delta; \varepsilon)$.

This lemma is a variant of Theorem 2.8(iv) of Polymath [4], and serves as replacement for Theorem 2.8(i).

Note: In Lemma 3, and in all subsequent lemmas, permissible (ω, γ, δ) are defined via strict inequalities. If we restrict the choice of (ω, γ, δ) to triples which satisfy these inequalities with room to spare (e.g. for some fixed ε_1 , we consider triples (ω, γ, δ) with $24\omega + 7\delta - 5\gamma + \varepsilon_1 < -2$ and $8\omega + 3\delta - \gamma + \varepsilon_1 < 0$), then ε can be chosen uniformly in (ω, γ, δ) . This means that we can ensure that the implied constant in the equidistribution estimate does not depend on the triple.

Lemma 4 (Polymath Type I(ii)). Suppose $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M and β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$ and $N(x) \leq x^{1/2}$. Suppose also that β has the Siegel-Walfisz property. Write $N(x) = x^\gamma$. Define the set of moduli

$$D_{IIb}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1/2-\gamma-2\omega-6\varepsilon-\delta} < u < x^{1/2-\gamma-2\omega-6\varepsilon} \end{array} \right\}.$$

Consider triples (ω, γ, δ) which satisfy the following inequalities:

$$\begin{aligned} 24\omega + 7\delta - 3\gamma &< -1, \\ 8\omega + 3\delta - \gamma &< 0. \end{aligned}$$

For such triples (ω, γ, δ) , convolution f has equidistribution properties for moduli in $D_{IIb}(x; \omega, \gamma, \delta; \varepsilon)$.

This lemma is a variant of Theorem 2.8(ii) of Polymath [4].

Lemma 5 (Baker-Irving Type I). Suppose $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M and β is a smooth coefficient sequence at scale N with $M(x)N(x) \asymp x$. Write $N(x) = x^\gamma$. Define the set of moduli

$$D_I(x; \omega, \gamma, \delta; \varepsilon) = \begin{cases} \{d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{\gamma-\delta-3\varepsilon} < r < x^{\gamma-3\varepsilon}\} & \text{if } \gamma \leq \frac{1}{2}, \\ \{d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{1-\gamma-\delta-3\varepsilon} < r < x^{1-\gamma-3\varepsilon}\} & \text{if } \gamma \in (\frac{1}{2}, \frac{1}{2} + 2\omega + \varepsilon], \\ [1, x^{1/2+2\omega}] \cap \mathbb{N} & \text{if } \gamma > \frac{1}{2} + 2\omega + \varepsilon. \end{cases}$$

Consider triples (ω, γ, δ) which satisfy the following inequalities:

$$\begin{aligned} 3\gamma - 12\omega - 3\delta &> 1 & \text{if } \gamma \leq \frac{1}{2}, \\ 68\omega + 14\delta &< 1 & \text{if } \gamma \in (\frac{1}{2}, \frac{1}{2} + 2\omega + \varepsilon]. \end{aligned}$$

For such triples (ω, γ, δ) , convolution f has equidistribution properties for moduli in $D_I(x; \omega, \gamma, \delta; \varepsilon)$.

This lemma is a variant of Lemma 5 of Baker and Irving [1].

Lemma 6 (S. Type I). Suppose $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M and β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$ and $N(x) \leq x^{1/2}$. Suppose also that β has the Siegel-Walfisz property. Write $N(x) = x^\gamma$. Define the set of moduli

$$D_{IIc}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \exists d_1 \mid r \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1-\gamma-6\varepsilon-\delta} d^{-1} < u < x^{1-\gamma-6\varepsilon} d^{-1} \\ \text{and } r^2 x^{2-\gamma-52\varepsilon-\delta} d^{-4} < d_1 < r^2 x^{2-\gamma-52\varepsilon} d^{-4} \end{array} \right\}.$$

Consider triples (ω, γ, δ) which satisfy the following inequalities:

$$\begin{aligned} 8\omega + 4\delta + 2\gamma &< 1, \\ 32\omega + 10\delta - \gamma &< 0, \\ 48\omega + 16\delta - 4\gamma &< -1. \end{aligned}$$

For such triples (ω, γ, δ) , convolution f has equidistribution properties for moduli in $D_{IIc}(x; \omega, \gamma, \delta; \varepsilon)$.

This lemma is a variant of Theorem 1 of [6].

Lemma 7 (Polymath Type III). Suppose $f(n; x) = (\alpha \star \psi_1 \star \psi_2 \star \psi_3)(n; x)$, where α is coefficient sequence at scale M and ψ_1, ψ_2 and ψ_3 are smooth coefficient sequences at scales N_1, N_2 and N_3 with

$$\begin{aligned} M(x)N_1(x)N_2(x)N_3(x) &\asymp x, \\ N_1(x)N_2(x), N_1(x)N_3(x), N_2(x)N_3(x) &\gg x^{1-\gamma}, \\ x^{1-2\gamma} &\ll N_1(x), N_2(x), N_3(x) \ll x^\gamma. \end{aligned}$$

Define the set of moduli

$$D_{III}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{1/3+4\delta/3-4\omega/3-\delta} < r < x^{1/3+4\delta/3-4\omega/3} \right\}.$$

Consider triples (ω, γ, δ) which satisfy the following inequality:

$$28\omega + 9\gamma + 8\delta < 4.$$

For such triples (ω, γ, δ) , convolution f has equidistribution properties for moduli in $D_{III}(x; \omega, \gamma, \delta; \varepsilon)$.

This lemma is a variant of Theorem 2.8(v) of Polymath [4].

3.3. Proofs. The original proofs of the lemmas and theorems referenced above span a total of over 50 pages. Since these proofs only need to be amended in a few specific places, it is not practical to write down full proofs of the new versions listed above. Instead we will just explain the particular changes we make. Our explanations are best read side-by-side with the original proofs.

3.3.1. Starting point. With the exception of Theorem 2.8(v) of Polymath [4], all results referenced above start their proofs in Section 5 of [4], and essentially proceed in exactly the same way from page 42 all the way until page 50. Although Polymath discusses i -tuply x^δ -densely divisible integers, pages 42 to 50 really just use that the relevant moduli m have a factorization $m = qr$ with $Nx^{-3\varepsilon-\delta_0} \ll r \ll Nx^{-3\varepsilon}$. As such, we begin by stating Polymath's Theorem 5.8 [4] with a suitably relaxed condition on the moduli. Since this statement involves exponential sums, we first need to introduce some more notation:

For $q \in \mathbb{N}$ and $n \in \mathbb{Z}$, we define $e_q(n) = \exp(2\pi i n/q)$. For $a, b \in \mathbb{Z}$, we set

$$e_q\left(\frac{a}{b}\right) = \begin{cases} e_q(\overline{a/b}) & \text{if } a/b \text{ is well-defined mod } q, \\ 0 & \text{otherwise,} \end{cases}$$

where $\overline{a/b}$ represents an integer n which satisfies $n \equiv a/b \pmod{q}$.

We will need to work with exponential sums of the following form:

Definition 8. For a given triple (ω, γ, δ) with $\gamma - 4\omega - \delta > 0$ and $\gamma \leq \frac{1}{2}$ and for $x > 0$, let $Z(\omega, \gamma, \delta; x)$ denote the set of $(Q, R, q_0, b_1, b_2, \ell)$ with the following properties:

- (i) $R \in (0, \infty)$ with $x^{\gamma-\delta-3\varepsilon} \leq R \leq x^{\gamma-2\varepsilon}$.
- (ii) $Q \in (0, \infty)$ with $x^{1/2-\varepsilon} \ll QR \ll x^{1/2+2\omega}$.
- (iii) $q_0 \in \mathbb{N}$ with $q_0 \ll Q$.
- (iv) $b_1, b_2 \in \mathbb{Z}$ with $(b_1 b_2, p) = 1$ for all $p \leq x$.
- (v) $\ell \in \mathbb{Z}$ with $1 \leq |\ell| \ll \frac{N}{R}$.

Consider some choice of sets $\mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) \subseteq [Q, 2Q] \cap \mathbb{N}$ and $\mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon) \subseteq [R, 2R] \cap \mathbb{N}$.

For a given $z = (Q, R, q_0, b_1, b_2, \ell) \in Z(\omega, \gamma, \delta; x)$, we then set $C(n) = 1_{\frac{b_1}{n} \equiv \frac{b_2}{n+\ell r} \pmod{q_0}}$ and define

$$\Sigma_{\mathcal{Q}, \mathcal{R}}(z) = \sum_{r \in \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon)} \sum_{\substack{q_1, q_2 \\ q_0 q_1, q_0 q_2 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) \\ (q_1, q_2) = 1 \\ q_0 q_1 r, q_0 q_2 r \text{ squarefree}}} \sum_{0 < |h| \leq \frac{x^\varepsilon R Q^2}{q_0 M}} \left| \sum_{\substack{n \\ (n, r q_0 q_1) = 1 \\ (n + \ell r, q_0 q_2) = 1}} C(n) \beta(n) \overline{\beta(n + \ell r)} \Phi_\ell(h, n, r, q_0, q_1, q_2) \right|,$$

$$\text{where } \Phi_\ell(h, n, r, q_0, q_1, q_2) = e_r \left(\frac{ah}{n q_0 q_1 q_2} \right) e_{q_0 q_1} \left(\frac{b_1 h}{n r q_2} \right) e_{q_2} \left(\frac{b_2 h}{(n + \ell r) r q_0 q_1} \right).$$

Now we are ready to summarize Theorem 5.8.

Lemma 8 (Alternative version of Theorem 5.8 of [4]). Suppose $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M and β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$. Suppose β has the Siegel-Walfisz property. Write $N(x) = x^\gamma$.

Consider some sets $\mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) \subseteq [Q, 2Q] \cap \mathbb{N}$ and $\mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon) \subseteq [R, 2R] \cap \mathbb{N}$.

Let $Z(\omega, \gamma, \delta; x)$ and $\Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell)$ be as defined in Definition 8.

Let $D_0 = \exp(\log(x)^{1/3})$ and suppose that $q \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ implies that $(q, p) = 1$ for $p \leq D_0$.

Suppose $D(x; \omega, \gamma, \delta; \varepsilon)$ is a set of moduli with

$$D(x; \omega, \gamma, \delta; \varepsilon) \subseteq \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists R \exists Q \text{ with } \log_2(Q) \in \mathbb{Z} \text{ and } \log_2(R) \in \mathbb{Z} \\ \text{and } x^{\gamma-\delta-3\varepsilon} \leq R \leq x^{\gamma-2\varepsilon} \text{ so that} \\ d = qr \text{ for some } q \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) \text{ and } r \in \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon), \\ \text{or } \prod_{p|d, p \leq D_0} p > \exp(\log^{2/3} x) \end{array} \right\}.$$

Consider triples (ω, γ, δ) with $\gamma - 4\omega - \delta > 0$ and $\gamma \leq \frac{1}{2}$. Suppose a collection of such triples satisfies

$$\sup_{(Q, R, q_0, b_1, b_2, \ell) \in Z(\omega, \gamma, \delta; x)} \Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell) \ll_\varepsilon x^{-2\varepsilon} Q^2 R N(q_0, \ell) q_0^{-2}.$$

Then convolution f has equidistribution properties for moduli in $D(x; \omega, \gamma, \delta; \varepsilon)$.

Proof. Only minor changes need to be made on pages 42 to 50 of [4]: Throughout, Polymath considers congruence classes a with $(a, p) = 1$ for $p \in I$, where I is some bounded set of integers (dependent on x). We simply choose $I = [1, x]$, and replace P_I by $\prod_{p \leq x} p$.

On page 43 it is observed that the contribution of d with $\prod_{p|d, p \leq D_0} p > \exp(\log^{2/3} x)$ can be bounded trivially. Hence we only think about $d = qr$ with $q \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ and $r \in \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon)$.

On page 44, Polymath notes that any i -tuply x^δ -densely divisible integer has a factorization $d = qr$ with $x^{\gamma-\delta-3\varepsilon} \leq r \leq x^{\gamma-3\varepsilon}$. Next prime factors $p \leq D_0$ are removed from q , which lowers the size of q by no more than x^ε . From then on, the proof works with sums over $q \in D_J^{(j)}(x^{\delta+o(1)}) \cap [Q, 2Q]$ and $r \in D_I^{(k)}(x^{\delta+o(1)}) \cap [R, 2R]$, which are $x^{\delta+o(1)}$ -densely divisible integers. We replace these by sums over $q \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ and $r \in \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon)$, where $x^{\gamma-\delta-3\varepsilon} \leq R \leq x^{\gamma-2\varepsilon}$. We already assumed that elements of $\mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ have no prime factors $p \leq D_0$, so we do not need to remove any factors. The condition $\log_2(Q), \log_2(R) \in \mathbb{Z}$ was included to highlight that we use dyadic decomposition and only need to consider $\ll \log(x)^2$ choices of Q and R . Finally, we may continue to assume that qr is squarefree.

On page 45, bound $RQ^2 \ll x$ is observed. Since $RQ \ll x^{1/2+2\omega}$ and $RQ^2 \ll x^{1+4\omega}/R \ll x^{1+4\omega-\gamma+\delta+3\varepsilon}$, this gives us the condition $\gamma - 4\omega - \delta > 0$. (Recall that we choose ε very small depending on (ω, γ, δ) . So we do not keep track of ε in the inequalities we record.)

On page 46, the Cauchy-Schwarz inequality is applied, giving us two copies of q , denoted by q_1 and q_2 . On page 48, the gcd $q_0 = (q_1, q_2)$ is introduced. However, later on page 49, relabeling occurs: q_1 is replaced by q_0q_1 and q_2 is replaced by q_0q_2 with $(q_1, q_2) = 1$. Thus from now on we sum over $q_0q_1 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ and $q_0q_2 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$. Having made these small changes, inequality (5.31) in Theorem 5.8 of [4] now instead reads $\Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell) \ll_\varepsilon x^{-2\varepsilon} Q^2 RN(q_0, \ell) q_0^{-2}$. $\square$

3.3.2. Proof of Lemma 3. The proof of Theorem 2.8(iv) of [4] continues on pages 51 to 54 and is very straightforward: we do not need to factorize moduli any further. However, it only applies to $\gamma \geq \frac{1}{2} - 2\omega$. Making a very minor change, we extend the proof to smaller values of γ , which serves as a replacement for Theorem 2.8(i). (Theorem 2.8(i) is a bit stronger, but this strength is not needed for our proofs, since γ close to $\frac{1}{2}$ behave well and do not constitute a critical parameter range.)

Proof of Lemma 3. We first apply Lemma 8 with a specific choice of $\mathcal{Q}$ and $\mathcal{R}$: Recall first that in Lemma 3 we work with moduli

$$D_{IIa}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{\gamma-\delta-3\varepsilon} < r < x^{\gamma-3\varepsilon} \right\}.$$

So $d \in D_{IIa}(x; \omega, \gamma, \delta; \varepsilon)$ implies $d = qr$ with $x^{\gamma-\delta-3\varepsilon} < r < x^{\gamma-3\varepsilon}$. Suppose now that $\prod_{p|d, p \leq D_0} p \leq \exp(\log^{2/3} x)$. Then $q = q^*s$ for some q^* with $(q^*, p) = 1$ for $p \leq D_0$ and some s with $s \ll x^\varepsilon$. Thus we can factorise d as $d = q^*(rs)$ where $x^{\gamma-\delta-3\varepsilon} < rs < x^{\gamma-2\varepsilon}$. Hence we can simply take

$$\begin{aligned} \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) &= \{q \in [Q, 2Q] \cap \mathbb{N} : (q, p) = 1 \text{ for } p \leq D_0\}, \\ \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon) &= [R, 2R] \cap \mathbb{N}. \end{aligned}$$

Applying Lemma 8, we must now show that

$$(3.2) \quad \sup_{(Q, R, q_0, b_1, b_2, \ell) \in Z(\omega, \gamma, \delta; x)} \Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell) \ll_\varepsilon x^{-2\varepsilon} Q^2 RN(q_0, \ell) q_0^{-2}.$$

Fixing a value of r , replacing condition $q_0q_1, q_0q_2 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ by $q_0q_1, q_0q_2 \in [Q, 2Q]$ (which increases the LHS), and replacing the absolute value signs in $\Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell)$ by constants c_{h, q_1, q_2} , we end up exactly with condition (5.33) on page 51 of [4]. The remaining proof of Theorem 2.8(iv) does not use any further factorization of moduli. Hence we can simply copy the arguments on pages 51 to 54 to show that (3.2) holds provided that

$$\begin{aligned} \frac{x^{1+12\omega+7\delta/2}}{N^{5/2}} &\ll_\varepsilon x^{-100\varepsilon}, \\ \frac{x^{8\omega+3\delta}}{N} &\ll_\varepsilon x^{-100\varepsilon}. \end{aligned}$$

(These inequalities can be found just below (5.36) on page 53.) At this point Polymath uses the assumption $\gamma > \frac{1}{2} - 2\omega - c$. However, we do not want to impose such a restriction and instead directly substitute $N = x^\gamma$. The inequalities listed above hold for sufficiently small ε , provided the following is true:

$$\begin{aligned} 24\omega + 7\delta - 5\gamma &< -2, \\ 8\omega + 3\delta - \gamma &< 0. \end{aligned}$$

Finally, recall that Lemma 8 had condition $4\omega + \delta - \gamma < 0$. However, this condition is already implied by $8\omega + 3\delta - \gamma < 0$, so we can discard it. $\square$

3.3.3. Exponential sum bounds. After arriving at Theorem 5.8, both Polymath [4] and Baker and Irving [1] apply exponential sum bounds which use the q -van der Corput method. (Theorem 2.8(iv), discussed above, is the one exception.) More specifically, they use the first inequality in Corollary 4.16 of [4]. However, using the q -van der Corput method necessitates the presence of another factor of convenient size in d . To keep conditions on the moduli simple, we would rather avoid this. Hence we will use a different exponential sum bound: the second inequality in Corollary 4.16 of [4], recorded below. This simpler (but potentially weaker) bound does not require us to consider further factorizations of d .

Lemma 9 (Corollary 4.16 of [4], inequality 2). *Let $N \geq 1$ and suppose $\psi_N(n)$ is a smooth coefficient sequence at scale N . Let $d_1, d_2 \in \mathbb{Z}$ be squarefree, and let $a, c_1, c_2, l_1, l_2 \in \mathbb{Z}$. Suppose $d \mid [d_1, d_2]$. Write $\delta_i = d_i / (d_1, d_2)$ and $\delta'_i = \delta_i / (d, \delta_i)$. Then for any $\varepsilon > 0$,*

$$\left| \sum_{n \equiv a(d)} \psi_N(n) e_{d_1} \left(\frac{c_1}{n + l_1} \right) e_{d_2} \left(\frac{c_2}{n + l_2} \right) \right| \ll_\varepsilon [d_1, d_2]^\varepsilon \left(\left(\frac{[d_1, d_2]}{d} \right)^{1/2} + \frac{1}{d} \frac{(c_1, \delta'_1)}{\delta'_1} \frac{(c_2, \delta'_2)}{\delta'_2} N \right).$$

Since this exponential sum bound uses gcds, the following simple lemma will also be useful:

Lemma 10 (Lemma 1.4 of [4]). *Let $q \in \mathbb{N}$ and $K \geq 1$. Then*

$$\sum_{1 \leq k \leq K} (k, q) \leq K\tau(q).$$

3.3.4. Proof of Lemma 4. The proof of Theorem 2.8(ii) of [4] involves a further factorization of $d = qr$ into $d = q_0 u v r$, which needs to be reflected in our choice of $\mathcal{D}$, $\mathcal{Q}$ and $\mathcal{R}$. In [4], the required size of u changes depending on q_0 , which is rather inconvenient. We thus carefully replace the range for u by a fixed interval, independent of q_0 . Additionally (as mentioned) above, we need to replace inequality 1 of Corollary 4.16 by inequality 2, so that we can simplify the factorization conditions further.

Proof of Lemma 4. Recall first that in Lemma 4 we work with moduli

$$D_{IIb}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1/2-\gamma-2\omega-6\varepsilon-\delta} < u < x^{1/2-\gamma-2\omega-6\varepsilon} \end{array} \right\}.$$

Hence we can take

$$\mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) = \left\{ q \in [Q, 2Q] \cap \mathbb{N} : \begin{array}{l} (q, p) = 1 \text{ for } p \leq D_0 \\ \text{and } \exists u \mid q \text{ with } x^{1/2-\gamma-2\omega-7\varepsilon-\delta} < u < x^{1/2-\gamma-2\omega-6\varepsilon} \end{array} \right\},$$

$$\mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon) = [R, 2R] \cap \mathbb{N}.$$

Applying Lemma 8, we must now show that

$$(3.3) \quad \sup_{(Q, R, q_0, b_1, b_2, \ell) \in Z(\omega, \gamma, \delta; x)} \Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell) \ll_\varepsilon x^{-2\varepsilon} Q^2 R N(q_0, \ell) q_0^{-2}.$$

We follow the steps on pages 57 to 59 of [4], making changes as follows: On page 57 Polymath uses that $q_0 q_1$ is $x^{\delta+o(1)}$ -densely divisible to deduce that q_1 has a factor u_1 between $q_0^{-1} x^{-\delta-2\varepsilon} Q/H$ and $x^{-2\varepsilon} Q/H$. We instead have $q_0 q_1 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon)$ and thus get $q_0 q_1 = uv$ with

$$x^{1/2-\gamma-2\omega-7\varepsilon-\delta} < u < x^{1/2-\gamma-2\omega-6\varepsilon}.$$

Recalling that $q_0 q_1 r$ is squarefree and writing $u_1 = u/(u, q_0)$ and $v_1 = v/(v, q_0)$, we then have $q_1 = u_1 v_1$ with $q_0^{-1} x^{1/2-\gamma-2\omega-7\varepsilon-\delta} < u_1 < x^{1/2-\gamma-2\omega-6\varepsilon}$. Thus we replace bound (5.39) on page 57 by

$$q_0^{-1} x^{1/2-\gamma-2\omega-7\varepsilon-\delta} \ll U \ll x^{1/2-\gamma-2\omega-6\varepsilon}.$$

Of course we still have $UV \asymp Q/q_0$. (Note: The condition $\gamma < 1/2 - 2\omega$ ensures that the upper bound on U is greater than 1, provided ε is sufficiently small.)

Continuing on, we can follow the exact same steps as Polymath to arrive at (5.42) on page 58. Summarizing their results up to this point, we now need to show the following:

(3.4)

$$\Upsilon_2 := \sum_{u_1 \asymp U} \sum_{q_2 \asymp Q/q_0} \sum_{\substack{v_1, v_2 \asymp V \\ (u_1 v_1 v_2, q_0 q_2) = 1}} \sum_{1 \leq |h_1|, |h_2| \leq \frac{x^\varepsilon R Q^2}{q_0 M}} \sup_{t(q_0)} \left| \sum_{n \equiv t(q_0)} \psi_N(n) \Phi_\ell(h_1, n, r, q_0, u_1 v_1, q_2) \overline{\Phi_\ell(h_2, n, r, q_0, u_1 v_2, q_2)} \right|$$

$$\ll_\varepsilon x^{-10\varepsilon} q_0^{-1} Q^2 V N.$$

At this point Polymath applies inequality 1 of Corollary 4.16. We instead apply inequality 2 (Lemma 9). Like Polymath, we use q_0 in the place of d , $rq_0u_1[v_1, v_2]$ in the place of d_1 and q_2 in the place of d_2 , and have $(c_2, \delta'_2)/\delta'_2 \leq 1$ and $(c_1, \delta'_1)/\delta'_1 \leq (c_1, r)/r = (h_1v_2 - h_2v_1, r)/r$. Hence

$$\begin{aligned} \Upsilon_2 &\ll_\varepsilon x^\varepsilon \sum_{u_1 \asymp U} \sum_{q_2 \asymp Q/q_0} \sum_{\substack{v_1, v_2 \asymp V \\ (u_1v_1v_2, q_0q_2)=1}} \sum_{1 \leq |h_1|, |h_2| \leq \frac{x^\varepsilon RQ^2}{q_0M}} \left((ru_1[v_1, v_2]q_2)^{1/2} + \frac{N}{q_0} \frac{(h_1v_2 - h_2v_1, r)}{r} \right) \\ &\ll_\varepsilon x^{2\varepsilon} q_0^{-3} U^{3/2} V^3 R^{5/2} Q^{11/2} M^{-2} + x^{4\varepsilon} \sum_{u_1 \asymp U} \sum_{q_2 \asymp Q/q_0} \sum_{|\Delta| \ll \frac{x^\varepsilon RQ^2 V}{q_0M}} \frac{N(\Delta, r)}{q_0 r} \sum_{v \asymp V} \sum_{1 \leq |h| \leq \frac{x^\varepsilon RQ^2}{q_0M}} \tau(\Delta - hv) \\ &\ll_\varepsilon x^{2\varepsilon} q_0^{-3} U^{3/2} V^3 R^{5/2} Q^{11/2} M^{-2} + x^{10\varepsilon} q_0^{-4} UV^2 RQ^5 NM^{-2} + x^{10\varepsilon} q_0^{-3} UVRQ^3 NM^{-1}. \end{aligned}$$

The bound (3.4) is therefore satisfied if the following inequalities are true:

$$\begin{aligned} q_0^{-2} U^{3/2} V^2 R^{5/2} Q^{7/2} M^{-2} N^{-1} &\ll_\varepsilon x^{-100\varepsilon}, \\ q_0^{-3} UVRQ^3 M^{-2} &\ll_\varepsilon x^{-100\varepsilon}, \\ q_0^{-2} URQM^{-1} &\ll_\varepsilon x^{-100\varepsilon}. \end{aligned}$$

Recall now that $RQ \ll x^{1/2+2\omega}$ and $R \gg x^{\gamma-3\varepsilon-\delta}$ and $q_0^{-1} x^{1/2-\gamma-2\omega-7\varepsilon-\delta} \ll U \ll x^{1/2-\gamma-2\omega-6\varepsilon}$ and $UV \asymp Q/q_0$. Hence it suffices to have (for sufficiently small ε):

$$\begin{aligned} -\frac{1}{2}(\frac{1}{2} - \gamma - 2\omega - \delta) + \frac{11}{2}(\frac{1}{2} + 2\omega) - 3(\gamma - \delta) + (\gamma - 2) &< 0, \\ 4(\frac{1}{2} + 2\omega) - 3(\gamma - \delta) - 2(1 - \gamma) &< 0, \\ (\frac{1}{2} - \gamma - 2\omega - 6\varepsilon) + (\frac{1}{2} + 2\omega) - (1 - \gamma) &< 0. \end{aligned}$$

Rearranging, we obtain our proposed conditions

$$\begin{aligned} 24\omega + 7\delta - 3\gamma &< -1, \\ 8\omega + 3\delta - \gamma &< 0. \end{aligned}$$

The condition $4\omega + \delta - \gamma < 0$ of Lemma 8 is again already implied by $8\omega + 3\delta - \gamma < 0$. $\square$

3.3.5. Proof of Lemma 5. Baker and Irving [1] use the assumption that coefficient sequence β is smooth and apply inequality 1 of Corollary 4.16 of [4] directly to $\Sigma_{Q, \mathcal{R}}$. We instead use inequality 2.

Proof of Lemma 5. We first consider $\gamma \leq \frac{1}{2}$. Recall that we work with moduli

$$D_I(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{\gamma-\delta-3\varepsilon} < r < x^{\gamma-3\varepsilon} \right\}.$$

Hence we can simply take

$$\begin{aligned} \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) &= \{q \in [Q, 2Q] \cap \mathbb{N} : (q, p) = 1 \text{ for } p \leq D_0\}, \\ \mathcal{R}(R; x, \omega, \gamma, \delta, \varepsilon) &= [R, 2R] \cap \mathbb{N}. \end{aligned}$$

Applying Lemma 8 with this choice of $\mathcal{Q}$ and $\mathcal{R}$, we must now show that

$$\sup_{(Q, R, q_0, b_1, b_2, \ell) \in Z(\omega, \gamma, \delta; x)} \Sigma_{\mathcal{Q}, \mathcal{R}}(Q, R, q_0, b_1, b_2, \ell) \ll_\varepsilon x^{-2\varepsilon} Q^2 RN(q_0, \ell) q_0^{-2}.$$

Like in the proof of Lemma 3 of [1], we observe that by fixing $r \in [R, 2R]$, this problem can be reduced to showing that

$$\begin{aligned} &\sum_{\substack{q_0 q_1, q_0 q_2 \in \mathcal{Q}(Q; x, \omega, \gamma, \delta, \varepsilon) \\ (q_1, q_2)=1 \\ q_0 q_1 r, q_0 q_2 r \text{ squarefree}}} \sum_{0 < |h| \leq \frac{x^\varepsilon RQ^2}{q_0M}} (q_0, \ell) \sup_{t(q_0)} \left| \sum_{n \equiv t(q_0)} \beta(n) \overline{\beta(n + \ell r)} e_{rq_0 q_1} \left(\frac{zh}{nq_2} \right) e_{q_2} \left(\frac{b_2 h}{(n + \ell r) r q_0 q_1} \right) \right| \\ &\ll_\varepsilon x^{-2\varepsilon} Q^2 N(q_0, \ell) q_0^{-2}, \end{aligned}$$

for a suitable choice of $z = z(a, b_1, r, q_0, q_1, q_2)$ with $(z, rq_0 q_1) = 1$.

We now apply Lemma 9 with $\beta(n) \overline{\beta(n + \ell r)}$ in the place of $\psi_N(n)$, t in the place of a , q_0 in the place of d , $rq_0 q_1$ in the place of d_1 and q_2 in the place of d_2 . Further, we take zh/q_2 in the place of c_1 , $b_2 h/(rq_0 q_1)$ in the place of c_2 , 0 in the place of l_1 and ℓr in the place of l_2 . This are the same choices as in [1], and thus

we also get $[d_1, d_2] = rq_0q_1q_2$ and $\delta'_1 = rq_1$ and $\delta'_2 = q_2$. Further, $(c_1, \delta'_1) = (h, rq_1)$ and $(c_2, \delta'_2) = (h, q_2)$. Hence we find that

$$\begin{aligned} & \sum_{\substack{q_1, q_2 \\ q_0q_1, q_0q_2 \in \mathcal{Q}(\mathcal{Q}; x, \omega, \gamma, \delta, \varepsilon) \\ (q_1, q_2)=1 \\ q_0q_1r, q_0q_2r \text{ squarefree}}} \sum_{0 < |h| \leq \frac{x^\varepsilon RQ^2}{q_0M}} (q_0, \ell) \sup_{t(q_0)} \left| \sum_{n \equiv t(q_0)} \beta(n) \overline{\beta(n + \ell r)} e_{rq_0q_1} \left(\frac{zh}{nq_2} \right) e_{q_2} \left(\frac{b_2 h}{(n + \ell r)rq_0q_1} \right) \right| \\ & \ll_\varepsilon \sum_{\substack{q_1, q_2 \\ q_0q_1, q_0q_2 \in \mathcal{Q}(\mathcal{Q}; x, \omega, \gamma, \delta, \varepsilon) \\ (q_1, q_2)=1 \\ q_0q_1r, q_0q_2r \text{ squarefree}}} \sum_{0 < |h| \leq \frac{x^\varepsilon RQ^2}{q_0M}} (q_0, \ell) (rq_0q_1q_2)^\varepsilon \left((rq_1q_2)^{1/2} + \frac{(h, rq_1q_2)}{rq_0q_1q_2} N \right) \\ & \ll_\varepsilon x^{3\varepsilon} \sum_{\substack{q_1, q_2 \\ q_0q_1, q_0q_2 \in \mathcal{Q}(\mathcal{Q}; x, \omega, \gamma, \delta, \varepsilon) \\ (q_1, q_2)=1 \\ q_0q_1r, q_0q_2r \text{ squarefree}}} (q_0, \ell) \left(\frac{R^{1/2}Q}{q_0} + \frac{q_0N}{RQ^2} \right) \left(\frac{RQ^2}{q_0M} \right) \ll_\varepsilon x^{3\varepsilon} \frac{(q_0, \ell)}{q_0^2} \left(\frac{R^{3/2}Q^5}{M} + \frac{Q^2N}{M} \right). \end{aligned}$$

To get the desired upper bound $\ll_\varepsilon x^{-2\varepsilon} Q^2 N(q_0, \ell) q_0^{-2}$, we thus simply need

$$\left(\frac{R^{3/2}Q^3}{MN} + \frac{1}{M} \right) \ll_\varepsilon x^{-5\varepsilon}.$$

Recalling $NM \asymp x$ and $RQ \ll x^{1/2+2\omega}$ and $R \gg x^{\gamma-\delta-3\varepsilon}$, and choosing ε sufficiently small (depending on (ω, γ, δ)), the inequality holds provided that

$$3(1/2 + 2\omega) - 3/2(\gamma - \delta) < 1.$$

This rearranges to condition $3\gamma - 12\omega - 3\delta > 1$ and concludes the treatment of $\gamma \leq \frac{1}{2}$.

Now we consider $\gamma \in [\frac{1}{2}, \frac{1}{2} + 2\omega + \varepsilon]$. If $24\omega + 7\delta - 5(1 - \gamma) < -2$ and $8\omega + 3\delta - (1 - \gamma) < 0$, we can apply our Lemma 3 (with the roles of α and β swapped and γ replaced by $1 - \gamma$). Since $\gamma \leq \frac{1}{2} + 2\omega + \varepsilon$, these inequalities are satisfied provided that

$$\min\left\{\frac{3}{5} - \frac{7}{5}\delta - \frac{24}{5}\omega, 1 - 8\omega - 3\delta\right\} > \frac{1}{2} + 2\omega.$$

Rearranging, the condition becomes

$$68\omega + 14\delta < 1.$$

Finally, if $\gamma > \frac{1}{2} + 2\omega + \varepsilon$, we can (like Baker and Irving) apply the results of pages 21 and 22 of Polymath [4]: for such large γ we have the desired equidistribution estimates for any $d \leq x^{1/2+2\omega}$, we do not need any factorization conditions.

Overall we thus have the desired equidistribution properties if $68\omega + 14\delta < 1$ and $3\gamma - 12\omega - 3\delta > 1$. $\square$

3.3.6. Proof of Lemma 6. Theorem 1 of [6] only applies to x^δ -smooth numbers. Again we extract much weaker factorization requirements from its proof to obtain Lemma 6. Fortunately, the original proof does not involve inequality 1 of Corollary 4.16 of [4], so although we do again change some factor sizes a bit (depending on q_0), calculations only change minimally.

Proof of Lemma 6. Recall that we work with moduli

$$D_{IIc}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \exists d_1 \mid r \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1-\gamma-6\varepsilon-\delta} d^{-1} < u < x^{1-\gamma-6\varepsilon} d^{-1} \\ \text{and } r^2 x^{2-\gamma-100\varepsilon-\delta} d^{-4} < d_1 < r^2 x^{2-\gamma-100\varepsilon} d^{-4} \end{array} \right\}.$$

We thus choose the following sets $\mathcal{Q}(Q, R; x, \omega, \gamma, \delta, \varepsilon)$ and $\mathcal{R}(Q, R; x, \omega, \gamma, \delta, \varepsilon)$:

$$\begin{aligned} \mathcal{Q}(Q, R; x, \omega, \gamma, \delta, \varepsilon) &= \left\{ q \in [Q, 2Q] \cap \mathbb{N} : \begin{array}{l} (q, p) = 1 \text{ for } p \leq D_0 \\ \text{and } \exists u \mid q \text{ with } x^{1-\gamma-6\varepsilon-\delta} (QR)^{-1} \ll u \ll x^{1-\gamma-6\varepsilon} (QR)^{-1} \end{array} \right\}, \\ \mathcal{R}(Q, R; x, \omega, \gamma, \delta, \varepsilon) &= \left\{ r \in [R, 2R] \cap \mathbb{N} : \exists d_1 \mid r \text{ with } x^{2-\gamma-52\varepsilon-\delta} (Q^4 R^2)^{-1} \ll d_1 \ll x^{2-\gamma-52\varepsilon} (Q^4 R^2)^{-1} \right\}. \end{aligned}$$

For any $d \in D_{IIc}(x; \omega, \gamma, \delta; \varepsilon)$ with $\prod_{p \mid d, p \leq D_0} p \leq \exp(\log^{2/3} x)$ there exist Q and R with $\log_2(Q) \in \mathbb{Z}$ and $\log_2(R) \in \mathbb{Z}$ and $x^{\gamma-3\varepsilon-\delta} \ll R \ll x^{\gamma-2\varepsilon}$ so that $d = qr$ for some $q \in \mathcal{Q}(Q, R; x, \omega, \gamma, \delta, \varepsilon)$ and some $r \in \mathcal{R}(Q, R; x, \omega, \gamma, \delta, \varepsilon)$.

We now follow the proof of Theorem 1 of [6] and make small alterations. The original proof starts on pages 40 to 43 in the appendix of [6]. These initial steps follow Polymath's proof of Theorem 2.8(ii)

rather closely, so we are already familiar with the necessary changes: In [6], $d \mid P(x^\delta)$ is used to deduce that $d = qr$ for some $r \in [R, 2R]$ with $x^{-4\epsilon-\delta}N \ll R \ll x^{-2\epsilon}N$. Although we might not have $d \mid P(x^\delta)$, the properties of $D(x; \omega, \gamma, \delta, \epsilon)$ ensure that we can also factorize like this. In particular, we can replace summation conditions like $(q_1 \asymp Q, r \asymp R \text{ and } q_1 r \mid P(x^\delta))$ by alternative conditions $(q_1 \in \mathcal{Q}(Q, R; x, \omega, \gamma, \delta, \epsilon) \text{ and } r \in \mathcal{R}(Q, R; x, \omega, \gamma, \delta, \epsilon) \text{ and } q_1 r \text{ squarefree})$. We also replace condition $(ab_1 b_2, P(x^\delta)) = 1$ by assumption $(ab_1 b_2, P(x)) = 1$.

On page 42 the original proof then uses $q_1 \mid P(x^\delta)$ to extract a factor of size U from q_1 . At this point, we instead have $q_0 q_1 \in \mathcal{Q}(Q, R; x, \omega, \gamma, \delta, \epsilon)$ and thus know that $q_0 q_1 = uv$ for some u with $x^{1-\gamma-6\epsilon-\delta}(QR)^{-1} \ll u \ll x^{1-\gamma-6\epsilon}(QR)^{-1}$. We then write $u_1 = u/(u, q_0)$ and $v_1 = v/(v, q_0)$ and get $q_1 = u_1 v_2$ for some $u_1 \asymp U$ with $q_0^{-1} x^{1-\gamma-6\epsilon-\delta}(QR)^{-1} \ll U \ll x^{1-\gamma-6\epsilon}(QR)^{-1}$. Recalling the notation $H = x^\epsilon R Q^2 / (q_0 M)$, this condition can be rewritten as

$$q_0^{-2} x^{-\delta-5\epsilon} Q/H \ll U \ll q_0^{-1} x^{-5\epsilon} Q/H$$

Hence we also get $x^{5\epsilon} H \ll V \ll q_0 x^{\delta+5\epsilon} H$. We then go to the main text. Our first change is to look at Lemma 3 (the summary of Polymath) and change the ranges of U, V and R in Definition 3 on page 10 accordingly.

On page 12 of [6], the proof of Theorem 1 then uses $r \mid P(x^\delta)$ to extract a factor d_1 from r , this time of size $D = N/(x^{50\epsilon} H^2)$. We instead recall that $r \in \mathcal{R}(R; x, \omega, \gamma, \delta, \epsilon)$ and hence get $r = d_1 r_1$ for some $d_1 \asymp \Delta$ with $x^{2-\gamma-52\epsilon-\delta}(Q^4 R^2)^{-1} \ll \Delta \ll x^{2-\gamma-52\epsilon}(Q^4 R^2)^{-1}$. This can be rewritten as

$$\frac{x^{-\delta} N}{q_0^2 x^{50\epsilon} H^2} \ll \Delta \ll \frac{N}{q_0^2 x^{50\epsilon} H^2}.$$

In particular, we have replaced $D \approx N/H^2$ by $D \approx q_0^{-2} N/H^2$, so that the value of D no longer depends on q_0 . On pages 12 to 17 of [6], the q -van der Corput process is carried out. Although we have slightly changed the ranges of R, U and Δ , the arguments on pages 12 to 17 can be copied with minimal alterations. We do need to be a bit careful when treating diagonal terms, which are covered by Lemma 7 on page 17 of [6]: because we increased V by a factor q_0 , various bounds now include an extra factor q_0^2 , and so we need to replace $\max\{\frac{x^{\delta+10\epsilon} H^3}{(w_2, m)}, \frac{H^4}{(w_2, m)}\}$ by $\max\{\frac{x^{\delta+10\epsilon} H^3 q_0^2}{(w_2, m)}, \frac{H^4}{(w_2, m)}\}$ in the upper bound on Σ_3 . This extra factor q_0^2 is carried through to later lemmas in the paper.

Definition 7 and Lemma 8 on page 20 of [6] summarize results up to this point. Due to extra factors q_0 , we need to make various amendments in this summary: $v_1, v_2 \ll x^{\delta+5\epsilon} H$ becomes $v_1, v_2 \ll q_0 x^{\delta+5\epsilon} H$, while $\frac{N}{x^{\delta+55\epsilon} H^2} \ll \Delta_1 \ll \frac{N}{x^{55\epsilon} H^2}$ becomes $\frac{N}{q_0^2 x^{\delta+55\epsilon} H^2} \ll \Delta_1 \ll \frac{N}{q_0^2 x^{55\epsilon} H^2}$. Further, $\frac{R Q^2 H}{q_0 (v_1, v_2) \Delta_1} \ll m \ll \frac{x^\delta R Q^2 H}{q_0 (v_1, v_2) \Delta_1}$ becomes $\frac{R Q^2 H}{q_0 (v_1, v_2) \Delta_1} \ll m \ll \frac{x^\delta R Q^2 H}{(v_1, v_2) \Delta_1}$ and finally $|\Lambda| \ll \frac{1}{w_1(v_1, v_2)} x^{\delta+5\epsilon} H^2$ becomes $|\Lambda| \ll q_0 \frac{1}{w_1(v_1, v_2)} x^{\delta+5\epsilon} H^2$. Fortunately, most of the remaining proof does not refer to specific parameter ranges for Δ_1, Λ and m , so from page 21 all the way until page 32 we can now proceed with minimal changes. However, it is convenient to notice that in Lemma 10 on page 23, we have the definition $\Delta^* = \min\{\frac{N}{\Lambda x^{5\epsilon}}, \Delta_1\}$. For our new choice of Δ_1 and for ϵ sufficiently small compared to δ , we always have $\frac{N}{\Lambda x^{5\epsilon}} \gg \frac{w_1(v_1, v_2) N}{q_0 x^{\delta+10\epsilon} H^2} \gg \frac{N}{q_0 x^{55\epsilon} H^2} \ll \Delta_1$ and can simply choose $\Delta^* = \Delta_1$.

To finish the proof, we now just need to amend the computations on page 33 of [6] with the extra factors q_0 we introduced. The equations (3.30), (3.31) and (3.32) which need to be verified now look as follows:

$$(3.5) \quad \frac{N \Delta_1}{m} \ll \min \left\{ \frac{1}{x^{\delta+10\epsilon} H^3 q_0^2}, \frac{1}{H^4} \right\} \frac{(v_1, v_2) q_0^2(q_0, \ell) N \Delta_1}{x^{\delta+131\epsilon} H^2},$$

$$(3.6) \quad N \ll \min \left\{ \frac{1}{x^{\delta+10\epsilon} H^3 q_0^2}, \frac{1}{H^4} \right\} \frac{(v_1, v_2) q_0^2(q_0, \ell) N \Delta_1}{x^{\delta+131\epsilon} H^2},$$

$$(3.7) \quad m \ll \min \left\{ \frac{1}{x^{\delta+10\epsilon} H^3 q_0^2}, \frac{1}{H^4} \right\} \frac{(v_1, v_2) q_0^2(q_0, \ell) N \Delta_1}{x^{\delta+131\epsilon} H^2}.$$

We begin with (3.5). This condition rearranges to

$$(3.8) \quad \left(\frac{x^{\delta+131\epsilon}}{(v_1, v_2) q_0^2(q_0, \ell)} \right) \frac{\max \{q_0^2 x^{\delta+10\epsilon} H^5, H^6\}}{m} \ll 1.$$

Recalling $m \gg \frac{R Q^2 H}{q_0 (v_1, v_2) \Delta_1} \gg \frac{q_0 x^{55\epsilon} R Q^2 H^3}{(v_1, v_2) N} \gg \frac{q_0 x^{54\epsilon} M H^4}{(v_1, v_2) N}$, we get that the LHS of (3.8) is bounded by

$$\left(\frac{x^{\delta+131\epsilon}}{(v_1, v_2) q_0^2(q_0, \ell)} \right) \frac{\max \{q_0^2 x^{\delta+10\epsilon} H^5, H^6\}}{m} \ll x^{-1+\delta+77\epsilon} N^2 \max \{x^{\delta+10\epsilon} H, H^2\} \ll x^{-1+8\omega+3\delta+100\epsilon} N^2.$$

This gives us the condition $8\omega + 4\delta + 2\gamma < 1$.

Next we consider (3.6). Rearranging, we find that this condition holds if

$$\frac{x^{\delta+131\varepsilon}}{q_0^2 \Delta_1} \max \{q_0^2 x^{\delta+10\varepsilon} H^5, H^6\} \ll 1.$$

Recalling that $\Delta_1 \gg \frac{N}{q_0^2 x^{\delta+55\varepsilon} H^2}$ and $H \ll \frac{x^{4\omega+\delta+7\varepsilon}}{q_0}$, we get that (3.6) holds if $32\omega + 10\delta - \gamma < 0$.

Finally, we consider (3.7). Recall that $m \ll \frac{x^\delta R Q^2 H}{(v_1, v_2) \Delta_1} \ll \frac{q_0 x^\delta M H^2}{(v_1, v_2) \Delta_1}$. So (3.7) is true if

$$\frac{x^{1+2\delta+131\varepsilon}}{N^2 \Delta_1^2} \max \{q_0^2 x^{\delta+10\varepsilon} H^7, H^8\} \ll 1.$$

Using $H \ll \frac{x^{4\omega+\delta+7\varepsilon}}{q_0}$, we see that (3.7) holds if the following two inequalities are satisfied:

$$(3.9) \quad \frac{x^{1+32\omega+10\delta+200\varepsilon}}{N^2} \left(\frac{x^{5\varepsilon} \Lambda}{\Delta_1 N} \right) \ll 1 \quad \text{and} \quad \frac{x^{1+32\omega+10\delta+200\varepsilon}}{N^2} \left(\frac{1}{\Delta_1^2} \right) \ll 1.$$

Using $\Lambda \ll x^{\delta+5\varepsilon} H^2$ and $\Delta_1 \gg \frac{N}{x^{\delta+55\varepsilon} H^2}$, it suffices to have $48\omega + 16\delta - 4\gamma < -1$. $\square$

3.3.7. Proof of Lemma 7. Theorem 2.8(v) of [4] requires that moduli d can be factorized as $d = qr$ with $r \asymp R$ for many different sizes of R . This is a bit inconvenient, and so we change the proof to work with a fixed value of R . Regarding notation, γ in this paper corresponds to $\frac{1}{2} - \sigma$ in [4].

Proof of Lemma 7. The proof of Theorem 2.8(v) begins on page 85 of [4], but its reduction to exponential sums does not need any factorization of moduli, and so we will mainly focus on the end of the proof, pages 90 to 93. Prior to page 90, factorization is mentioned once, below (7.17). There we have $b = \prod_{p|b_1 b_2 b_3} p$ and it is observed that $q = bd$ implies $d \in \mathcal{D}_I(bx^\delta)$. We instead recall that in Lemma 3 we work with

$$D_{III}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ q \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid q \text{ with } x^{1/3+4\delta/3-4\omega/3-\delta} < r < x^{1/3+4\delta/3-4\omega/3} \right\}.$$

We thus define the set

$$\mathcal{D}'(b; x; \omega, \gamma, \delta) = \left\{ d \in \mathbb{N} : \exists r \mid d \text{ with } b^{-1} x^{1/3+4\delta/3-4\omega/3-\delta} < r < x^{1/3+4\delta/3-4\omega/3} \right\}$$

Consider a squarefree $q \in D(x; \omega, \gamma, \delta; \varepsilon)$. We have $q = rs$ for some r with $x^{1/3+4\delta/3-4\omega/3-\delta} < r < x^{1/3+4\delta/3-4\omega/3}$. If also $q = bd$, we take $r' = r/(r, b)$ and $s' = s/(s, b)$, so that $d = r's'$ with $b^{-1} x^{1/3+4\delta/3-4\omega/3-\delta} < r' < x^{1/3+4\delta/3-4\omega/3}$. Hence we replace condition $d \in \mathcal{D}_I(bx^\delta)$ by condition $d \in \mathcal{D}'(b; x; \omega, \gamma, \delta)$.

Summarizing results up to page 89, the goal of pages 90 to 93 is as follows: We must show that

$$(3.10) \quad \Sigma'_2 = \frac{N_1 N_2 N_3}{Q^2} \sum_{(b_1, b_2, b_3)} \frac{b_1^b b_2^b b_3^b}{b^2} T(b_1, b_2, b_3) \ll_\varepsilon x^{1-3\varepsilon}$$

$$\text{where } T(b_1, b_2, b_3) = \sum_{0 < |\ell| \leq \frac{x^{3\varepsilon/2} Q^3}{b_1 b_2 b_3 N}} \left| \sum_{\substack{d \asymp Q/b \\ d \in \mathcal{D}'(b; x; \omega, \gamma, \delta) \\ d \text{ squarefree} \\ (b\ell, d)=1}} \eta'_{bd} \sum_{(m, bd)=1} \alpha(m) \text{Kl}_3 \left(\frac{a\ell b_1 b_2 b_3}{b^3 m}; d \right) \right|$$

Here $x^{1/2-\varepsilon} \ll Q \ll x^{1/2+2\omega}$, η'_{bd} is some sequence of coefficients supported on short interval $\mathcal{I}(Q)$ and $b = \prod_{p|b_1 b_2 b_3} p$ for some $b_i \in \mathbb{N}$ with $b \ll Q$ and $b_1 b_2 b_3 \ll x^{3\varepsilon/2} Q^3 / N$. Further, b_i^b is the largest squarefree divisor of b_i and $b = (b_1 b_2 b_3)^b$.

On page 90, Polymath begins by using $d \in \mathcal{D}_I(bx^\delta)$ to deduce that $d = rs$ with $(bx^\delta)^{-1} S \leq s \leq S$. Of course, we can use $d \in \mathcal{D}'(b; x; \omega, \gamma, \delta)$ to deduce the same with $S = x^{1/3+4\delta/3-4\omega/3}$. This choice of S satisfies the requirement $1 \leq S \leq x^\delta Q/2$ provided that $4\omega - 4\delta < 1$ and $2\delta - 8\omega < 1$.

Fortunately, Polymath initially proceeds without specifying their value of S . As such, we can simply copy their arguments on pages 90 and 91. Noting that $H_b = \frac{x^{3\varepsilon/2} H}{b_1 b_2 b_3} = \frac{x^{3\varepsilon/2} Q^3}{b_1 b_2 b_3 N}$ where $N = N_1 N_2 N_3$, we get $|T(b_1, b_2, b_3)|^2 \leq T_1 T_2$ with $T_1 \ll_\varepsilon \frac{x^{3\varepsilon} Q^3}{b_1 b_2 b_3 N}$ and

$$T_2 = T'_2 + T''_2 \ll_\varepsilon \frac{x^{3\varepsilon} M Q^4 S}{b b_1 b_2 b_3 N} + \frac{x^{3\varepsilon} M^2 Q^4 S^{1/2}}{b b_1 b_2 b_3 N} + \frac{x^{3\varepsilon} x^{\delta/2} M^2 Q^3}{b^{5/2} S^{1/2}}.$$

Multiplying T_1 and T_2 together, substituting $S = x^{1/3+4\delta/3-4\omega/3}$ and $Q \ll x^{1/2+2\omega}$ and recalling that $x^{3/2-3\gamma/2} \ll N \ll x^{3\gamma}$, we now have

$$\begin{aligned} |T(b_1, b_2, b_3)| &\ll_\varepsilon \frac{x^{3\varepsilon} M^{1/2} Q^{7/2} S^{1/2}}{b^{1/2} b_1 b_2 b_3 N} + \frac{x^{3\varepsilon} M Q^{7/2} S^{1/4}}{b^{1/2} b_1 b_2 b_3 N} + \frac{x^{3\varepsilon} x^{\delta/4} M Q^3}{b^{5/4} (b_1 b_2 b_3)^{1/2} S^{1/4} N^{1/2}} \\ &\ll_\varepsilon \left(\frac{x^{3\varepsilon} Q^2}{N_1 N_2 N_3} \right) \left(\frac{x^{1/2+3\omega+3\gamma/4} S^{1/2}}{b^{1/2} b_1 b_2 b_3} + \frac{x^{1/4+3\omega+3\gamma/2} S^{1/4}}{b^{1/2} b_1 b_2 b_3} + \frac{x^{\delta/4+3/4+2\omega+3\gamma/4}}{b^{5/4} (b_1 b_2 b_3)^{1/2} S^{1/4}} \right) \\ &\ll_\varepsilon \left(\frac{x^{3\varepsilon} Q^2}{N_1 N_2 N_3} \right) \left(\frac{x^{2/3+7\omega/3+3\gamma/4+2\delta/3}}{b (b_1 b_2 b_3)^{1/2}} + \frac{x^{1/3+8\omega/3+3\gamma/2+\delta/3}}{b (b_1 b_2 b_3)^{1/2}} + \frac{x^{7\omega/3+3\gamma/4+2/3-\delta/12}}{b (b_1 b_2 b_3)^{1/2}} \right). \end{aligned}$$

Substituting this upper bound on $|T(b_1, b_2, b_3)|$ back into (3.10), we get

$$\Sigma'_2 \ll_\varepsilon x^{3\varepsilon} \left(x^{2/3+7\omega/3+3\gamma/4+2\delta/3} + x^{1/3+8\omega/3+3\gamma/2+\delta/3} + x^{7\omega/3+3\gamma/4-5/6-\delta/12} \right) \sum_{(b_1, b_2, b_3)} \frac{b_1^b b_2^b b_3^b}{b^3 (b_1 b_2 b_3)^{1/2}}.$$

But by Lemma 7.5 of [4], the inner sum over (b_1, b_2, b_3) is convergent. Therefore we have $\Sigma'_2 \ll_\varepsilon x^{1-3\varepsilon}$ provided the following three inequalities hold:

$$\begin{aligned} 2/3 + 7\omega/3 + 3\gamma/4 + 2\delta/3 &< 1, \\ 1/3 + 8\omega/3 + 3\gamma/2 + \delta/3 &< 1, \\ 7\omega/3 + 3\gamma/4 + 2/3 - \delta/12 &< 1. \end{aligned}$$

The second and third inequality are already implied by the first, and so this leaves us with the condition $28\omega + 9\gamma + 8\delta < 4$. $\square$

4. FINDING PERMISSIBLE PARAMETERS

In the previous sections we first derived a version of the GPY sieve with coefficients supported on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, conditional on there existing a minorant $\rho(n; x)$ for the prime indicator function which has suitable equidistribution properties for moduli produced by $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. We then derived a number of equidistribution estimates which work well for moduli with large x^δ -smooth factors. Now we need to put these ingredients together to choose good values for δ , $\underline{A}$ and $\underline{B}$ and construct a suitable $\rho(n; x)$.

4.1. Harman's sieve. We begin with a Harman's sieve construction.

Definition 9 (Decomposition). *Let $\epsilon = 10^{-10}$ and let $\xi_1, \xi_2, \xi_3 \in (0, 1)$. Then $\mathcal{H}(\xi_1, \xi_2, \xi_3)$ denotes the set of functions $f : \mathbb{N} \times (1, \infty) \rightarrow \mathbb{C}$ which satisfy one of the following three conditions:*

- (I) *$f(n; x) = (\alpha \star \beta)(n; x)$, where α is a coefficient sequence at scale M and β is a smooth coefficient sequence at scale N with $M(x)N(x) \asymp x$. Additionally, $N(x) = x^{\gamma(x)}$ with $\gamma(x) \geq \xi_1 - \epsilon$.*
- (II) *$f(n; x) = (\alpha \star \beta)(n; x)$, where α is a coefficient sequence at scale M , β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$, and α and β both have the Siegel–Walfisz property. Additionally, $N = x^{\gamma(x)}$ with $\xi_2 - \epsilon \leq \gamma(x) \leq 1 - \xi_2 + \epsilon$.*
- (III) *$f(n; x) = (\alpha \star \psi_1 \star \psi_2 \star \psi_3)(n; x)$, where α is a coefficient sequence at scale M and ψ_1, ψ_2 and ψ_3 are smooth coefficient sequences at scales N_1, N_2 and N_3 with $M(x)N_1(x)N_2(x)N_3(x) \asymp x$. Additionally, $x^{1-2\xi_3-\epsilon} \leq N_1(x), N_2(x), N_3(x) \leq x^{\xi_3+\epsilon}$ and $N_i(x)N_j(x) \geq x^{1-\xi_3-\epsilon}$ for $i \neq j$.*

We will now use $\mathcal{H}(\xi_1, \xi_2, \xi_3)$ to construct a minorant for the primes.

Proposition 2. *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1, let $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ be as given in Definition 2 and let $\mathcal{H}(\xi_1, \xi_2, \xi_3)$ be as given in Definition 9.*

Suppose $\xi_1, \xi_2, \xi_3 \in (0, 1)$ satisfy all of the following inequalities:

$$\begin{aligned} 2\xi_1 + 3\xi_2 &< 2, \\ \xi_2 &\leq \xi_3, \\ \xi_1 + 9\xi_2 &< 4, \\ 2\xi_1 + \xi_2 &> 1, \\ 17\xi_2 &< 7. \end{aligned}$$

Suppose further that for every $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$, $x > 1$, $\varepsilon_0 > 0$, $A > 0$ and $a \in \mathbb{Z}$ with $(a, P(x)) = 1$,

$$(4.1) \quad \sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0) \\ q \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(q)}} f(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x, 2x] \\ (n, q) = 1}} f(n; x) \right| \ll_{A, \varepsilon_0} \frac{x}{\log(x)^A}.$$

We then write $p_j = x^{\alpha_j}$ and define the minorant

$$\rho(n; x) = 1_{\mathbb{P}}(n) - \sum_{\substack{n=p_1 p_2 p_3 p_4 p_5 \\ 1-2\xi_2 < \alpha_4 < \alpha_3 < \alpha_2 < \alpha_1 < \xi_2 \\ \alpha_1 + \alpha_2 < \xi_2 \\ \alpha_2 + \alpha_3 + \alpha_4 > 1 - \xi_2 \\ \alpha_5 > \alpha_4}} 1 - \sum_{\substack{n=p_2 p_3 p_4 p_5 p_6 \\ 1-2\xi_2 < \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6 < 8\xi_2 - 3 \\ \alpha_2, \alpha_4 > \alpha_3 \\ \alpha_2 + \alpha_4 < \xi_2 \\ \alpha_3 + \alpha_4 + \alpha_5 > 1 - \xi_2 \\ \alpha_6 > \alpha_5}} 1.$$

Furthermore, we choose the constants

$$c_1 = \int_{1-2\xi_2}^{\xi_2} \int_{1-2\xi_2}^{\min\{\alpha_1, \xi_2 - \alpha_1\}} \int_{1-2\xi_2}^{\alpha_2} \int_{\max\{1-2\xi_2, 1-\xi_2 - \alpha_2 - \alpha_3\}}^{\min\{\alpha_3, (1-\alpha_1 - \alpha_2 - \alpha_3)/2\}} \frac{1}{\alpha_1 \alpha_2 \alpha_3 \alpha_4 (1 - \sum_{i=1}^4 \alpha_i)} d\alpha_1 d\alpha_2 d\alpha_3 d\alpha_4 \\ + \int_{1-2\xi_2}^{8\xi_2 - 3} \int_{1-2\xi_2}^{\min\{8\xi_2 - 3, \alpha_2\}} \int_{\max\{\alpha_3, 1-2\xi_2\}}^{\min\{8\xi_2 - 3, \xi_2 - \alpha_2\}} \int_{\max\{1-2\xi_2, 1-\xi_2 - \alpha_3 - \alpha_4\}}^{\min\{8\xi_2 - 3, (1 - \sum_{i=2}^4 \alpha_i)/2\}} \frac{1}{\prod_{i=2}^5 \alpha_i (1 - \sum_{i=2}^5 \alpha_i)} d\alpha_2 d\alpha_3 d\alpha_4 d\alpha_5, \\ c_2 = \begin{cases} 24 & \text{if } \xi_2 > \frac{4}{10}, \\ 0 & \text{if } \xi_2 \leq \frac{4}{10}. \end{cases}$$

Then $\rho(n; x)$ has the following properties:

- (1) Prime minorant: $-c_2 \leq \rho(n; x) \leq 1_{\mathbb{P}}(n)$ for all large x and $n \in [x, 2x]$.
- (2) ρ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. (See Definition 3.)
- (3) If $\rho(n; x) \neq 0$, then all prime factors of n exceed $x^{1-2\xi_2}$.
- (4) The sum of $\rho(n; x)$ over $[x, 2x]$ satisfies $\sum_{n \in [x, 2x]} \rho(n; x) = (1 - c_1 + o(1)) \frac{x}{\log(x)}$.

(Note: If $\xi_2 \leq \frac{4}{10}$, then $\rho(n; x)$ is simply $1_{\mathbb{P}}(n)$.)

Proof. The construction of $\rho(n; x)$ is essentially the same as in Baker and Irving's work [1] and in the author's paper [6]. We have simply replaced specific numeric values by variables. However, along the way numerous inequalities need to be satisfied, so to justify this choice of $\rho(n; x)$, we look a little more closely at the proofs in Section 4 of [6]: The construction begins on page 35 with the statement of various equidistribution estimates. We replace the values 0.33856, 0.40481, 0.59519, 0.19, 0.405 and 0.595, which are used in [6], by $\xi_1, \xi_2, 1 - \xi_2, 1 - 2\xi_3, \xi_3$ and $1 - \xi_3$, respectively. In Lemma 19 on page 35 we then instead have $\alpha = \xi_2, \beta = 1 - 2\xi_2 = \lambda$ and $M = x^{1-\xi_1}$ and $\zeta = 1 - \xi_1 - \xi_2$. In particular, 0.19038 is replaced by $1 - 2\xi_2$ and 0.25663 is replaced by $1 - \xi_1 - \xi_2$. Using Buchstab decompositions as on page 36 of [6], and making the assumption

$$2\zeta = 2(1 - \xi_1 - \xi_2) > \xi_2,$$

we then arrive at

$$(4.2) \quad 1_{\mathbb{P}}(n) = \theta_0(n; x) + \sum_{\substack{n=p_1 p_2 n_3 \\ 1-2\xi_2 \leq \alpha_2 < \alpha_1 < \xi_2 \\ \alpha_1 + \alpha_2 > 1 - \xi_2 \\ \alpha_2 > 1 - \xi_1 - \xi_2}} \psi(n_3, p_2) + \sum_{\substack{n=p_1 p_2 p_3 p_4 n_5 \\ 1-2\xi_2 \leq \alpha_4 < \alpha_3 < \alpha_2 < \alpha_1 < \xi_2 \\ \alpha_1 + \alpha_2 < \xi_2 \\ \alpha_3 < 1 - \xi_1 - \xi_2}} \psi(n_5, p_4) - \sum_{\substack{n=p_1 p_2 p_3 n_4 \\ 1-2\xi_2 \leq \alpha_3 < \alpha_2 < \alpha_1 < \xi_2 \\ \alpha_1 + \alpha_2 > 1 - \xi_2 \\ \alpha_2 < 1 - \xi_1 - \xi_2}} \psi(n_4, p_3),$$

where $\theta_0(n; x)$ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$.

For $\xi_1 = 0.33856$ and $\xi_2 = 0.40481$, the first sum in (4.2) is treated in Lemma 20 on page 36 of [6]. There it is shown that this sum also has the desired equidistribution properties. For this proof to work for our more general parameters ξ_1, ξ_2 and ξ_3 , we need the following: At the start of the proof of Lemma 20 we use that n_3 must be prime. Since $\alpha_1 + \alpha_2 > 1 - \xi_2$ and n_3 not prime would imply $n_3 > x^{2(1-\xi_1-x_2)}$ this is ensured by assumption $2(1 - \xi_1 - x_2) > \xi_2$. Next, in the fourth paragraph on page 37, Polymath's Type III equidistribution estimate is implied. To make sure this is possible, we have the important assumption

$$\xi_2 \leq \xi_3.$$

A paragraph further, $\mu_2 + \mu_3 < \xi_2$ is treated. There $1 - \xi_1 - \xi_2/2 > \xi_2$ is used to get $\mu_4, \mu_5 > 1 - 2\xi_2$. (For the meaning of μ_i , see [6]. We skip this discussion here, as it is not relevant for the inequality check.) $\mu_1 + (\mu_6 + \dots + \mu_j) < 1 - \xi_1 - \xi_2$ is then deduced from $1 - 4(1 - 2\xi_2) < 1 - \xi_1 - \xi_2$, which rearranges to

$$\xi_1 + 9\xi_2 < 4.$$

The inequalities $2\xi_1 + 3\xi_2 < 2$ and $\xi_2 < \xi_3$ and $\xi_1 + 9\xi_2 < 4$ are all we need for the proof of Lemma 20 to work. Moving on, the second sum in (4.2) is treated by Lemma 21 on page 37.

For the proof of Lemma 21 to work with general ξ_1, ξ_2 and ξ_3 , we need the following: First, the second sum is split up depending on whether $\alpha_2 + \alpha_3 + \alpha_4 \leq 1 - \xi_2$ or $\alpha_2 + \alpha_3 + \alpha_4 > 1 - \xi_2$. In the first case,

we need $3(1 - 2\xi_2) > \xi_2$ to deduce that the sum has the desired equidistribution properties. Here we get the additional assumption $7\xi_2 < 3$.

In the second case, we use $\xi_2/2 < 1 - \xi_1 - \xi_2$ to drop the assumption $\alpha_3 < 1 - \xi_1 - \xi_2$, and $\alpha_2 + \alpha_3 + \alpha_4 > 1 - \xi_2$ means that we need $(4/3)(1 - \xi_2) + 2(1 - 2\xi_2) > 1$ to deduce that n_5 is prime. The condition $(4/3)(1 - \xi_2) + 2(1 - 2\xi_2) > 1$ rearranges to $16\xi_2 < 7$, which is already implied by $7\xi_2 < 3$. Hence we are left with the sum

$$(4.3) \quad \Gamma_2(n; x) = \sum_{\substack{n=p_1 p_2 p_3 p_4 p_5 \\ 1-2\xi_2 < \alpha_4 < \alpha_3 < \alpha_2 < \alpha_1 < \xi_2 \\ \alpha_1 + \alpha_2 < \xi_2 \\ \alpha_2 + \alpha_3 + \alpha_4 > 1 - \xi_2 \\ \alpha_5 > \alpha_4}} 1,$$

for which we do not have good equidistribution estimates. It thus needs to be subtracted from $1_{\mathbb{P}}(n)$. Using the prime number theorem for short intervals, we find that $\sum_{n \in [x, 2x]} \Gamma_2(n; x)$ equals

$$\frac{x(1 + o(1))}{\log(x)} \int_{1-2\xi_2}^{\xi_2} \int_{1-2\xi_2}^{\min\{\alpha_1, \xi_2 - \alpha_1\}} \int_{1-2\xi_2}^{\alpha_2} \int_{\max\{1-2\xi_2, 1-\xi_2-\alpha_2-\alpha_3\}}^{\min\{\alpha_3, (1-\alpha_1-\alpha_2-\alpha_3)/2\}} \frac{1}{\alpha_1 \alpha_2 \alpha_3 \alpha_4 (1 - \sum_{i=1}^4 \alpha_i)} d\alpha_1 d\alpha_2 d\alpha_3 d\alpha_4.$$

Finally, we need to look at the third sum in (4.2). It is treated by Lemma 22 on page 38 of [6]. The proof first needs $(1 - \xi_2) + 3(1 - 2\xi_2) > 1$ to deduce that n_4 is prime. This is implied by $7\xi_2 < 3$. Next, $(1 - \xi_2)/2 > 1 - \xi_1 - \xi_2$ is needed to drop condition $\alpha_2 < \alpha_1$. This condition rearranges to

$$2\xi_1 + \xi_2 > 1.$$

Further, we have $1 - 3(1 - 2\xi_2) < 1 - \xi_2$ (since $7\xi_2 < 3$), and this inequality is used to drop condition $\alpha_1 < 1 - \xi_2$. At that point the proof of Lemma 22 splits the sum into two parts, Υ_1 and Υ_2 . Υ_2 has good equidistribution properties, while reversal of roles is applied to Υ_1 . We end up with

$$\Upsilon_3(n; x) = \sum_{\substack{n=p_2 p_3 p_4 p_5 p_6 \\ 1-2\xi_2 \leq \alpha_3 < \alpha_2 < \xi_2 \\ \alpha_3 + \alpha_4 < \xi_2 \\ \alpha_2 < 1 - \xi_1 - \xi_2 \\ \alpha_4 \geq \alpha_3 \\ 1-2\xi_2 \leq \alpha_5 \leq \alpha_6}} 1,$$

where $6(1 - 2\xi_2) > 1$ was needed to get the prime p_6 . This rearranges to condition $12\xi_2 < 5$. Note that $12\xi_2 < 5$ implies $7\xi_2 < 3$. Next we observe that $\alpha_i > 1 - 2\xi_2$ for all i implies that also $\alpha_i < 1 - 4(1 - 2\xi_2) = 8\xi_2 - 3$ for all i . Furthermore, $8\xi_2 - 3 < \xi_2$ since $7\xi_2 < 3$. So condition $\alpha_i < \xi_2$ can be replaced by $\alpha_i < 8\xi_2 - 3$. Finally, the proof of Lemma 22 also requires $2(8\xi_2 - 3) < 1 - \xi_2$. We need

$$17\xi_2 < 7$$

to get that Υ_3 is the sum of a function with the desired equidistribution properties and of $\Gamma_2^*(n; x)$, where

$$(4.4) \quad \Gamma_2^*(n; x) = \sum_{\substack{n=p_2 p_3 p_4 p_5 p_6 \\ 1-2\xi_2 \leq \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6 \leq 8\xi_2 - 3 \\ \alpha_2, \alpha_4 > \alpha_3 \\ \alpha_2 + \alpha_4 < \xi_2 \\ \alpha_3 + \alpha_4 + \alpha_5 > 1 - \xi_2 \\ \alpha_6 > \alpha_5}} 1.$$

Using standard techniques, we find that $\sum_{n \in [x, 2x]} \Gamma_2^*(n; x)$ is asymptotically equal to

$$\frac{x(1 + o(1))}{\log(x)} \int_{1-2\xi_2}^{8\xi_2 - 3} \int_{1-2\xi_2}^{\min\{8\xi_2 - 3, \alpha_2\}} \int_{\max\{\alpha_3, 1-2\xi_2\}}^{\min\{8\xi_2 - 3, \xi_2 - \alpha_2\}} \int_{\max\{1-2\xi_2, 1-\xi_2-\alpha_3-\alpha_4\}}^{\min\{8\xi_2 - 3, (1 - \sum_{i=2}^4 \alpha_i)/2\}} \frac{1}{\prod_{i=2}^5 \alpha_i (1 - \sum_{i=2}^5 \alpha_i)} d\alpha_2 d\alpha_3 d\alpha_4 d\alpha_5.$$

Overall, this tells us that $\rho(n; x)$ has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, and that $\sum_{n \in [x, 2x]} \rho(n; x) = (1 - c_1 + o(1)) \frac{x}{\log(x)}$ for the given value of c_1 . Furthermore, clearly $\rho(n; x) \leq 1_{\mathbb{P}}(n)$ and $\rho(n; x) \neq 0$ implies that all prime factors of n exceed $x^{1-2\xi_2}$. Now we need to think about c_2 .

If $\xi_2 \leq \frac{4}{10}$, we note that $1 - 2\xi_2 \geq \frac{1}{5}$ and hence $\alpha_i > \frac{1}{5}$ for all i in both $\Gamma_2(n; x)$ and $\Gamma_2^*(n; x)$. This means that $\Gamma_2(n; x)$ and $\Gamma_2^*(n; x)$ are simply empty sums and $\rho(n; x) = 1_{\mathbb{P}}(n)$. We can take $c_2 = 0$. If $\xi_2 > \frac{4}{10}$ we need to be more careful. Looking at $\Gamma_2(n; x)$, defined in (4.3), we only get a negative contribution if $n = p'_1 p'_2 p'_3 p'_4 p'_5$ for some primes p'_i with $p'_1 > \dots > p'_5$. Further, p_4 needs to be the smallest prime, so there are only 4 choices for p_5 . Since $p_1 > p_2 > p_3 > p_4$, the choice of p_5 immediately also determines the values of p_1, p_2 and p_3 . No n can be counted more than 4 times. Thus $\Gamma_2(n; x) \geq -4$. Similarly, looking at $\Gamma_2^*(n; x)$, defined in (4.4), we only get a negative contribution if $n = p'_1 p'_2 p'_3 p'_4 p'_5$ for some primes p'_i with $p'_1 > \dots > p'_5$. There are at most $\binom{5}{3} = 10$ choices for $\{p_2, p_3, p_4\}$, and since we need $p_2 > p_3$ and

$p_4 > p_3$ and $p_6 > p_5$, any n can be counted at most 20 times. We get $\Gamma_2(n; x) \geq -20$ and overall have $\rho(n; x) \geq -24$. This gives us choice $c_2 = 24$. $\square$

4.2. Partition lemmas. In Proposition 2 we chose parameters (ξ_1, ξ_2, ξ_3) and constructed a prime minorant $\rho(n; x)$ which has equidistribution properties for moduli corresponding to $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ provided that every $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$ has equidistribution properties for moduli in $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$. We now need to translate the equidistribution estimates proved in Section 3 into conditions on (ξ_1, ξ_2, ξ_3) and $(\delta, \underline{A}, \underline{B})$ which ensure that $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$ equidistributes for moduli in $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$. We will do this by thinking about the following sets:

Definition 10. For given $B_1, B_2, \delta \in (0, \infty)$ and given $m_1, m_2 \in \mathbb{N} \cup \{0\}$, we set

$$\Xi(B_1, B_2, m_1, m_2, \delta) = \left\{ (y_1, \dots, y_{m_1+m_2}) \in [\delta, 1]^{m_1+m_2} : \sum_{i=1}^{m_1} y_i \leq B_1 \text{ and } \sum_{i=m_1+1}^{m_1+m_2} y_i \leq B_2 \right\}.$$

Properties of $\Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ will be used to factorize moduli in $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$.

4.2.1. 2 factors. We begin by asking what conditions on $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ ensure that $q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ has a factor in some fixed interval $[x^a, x^b]$. We make the following simple observation:

Lemma 11. Let $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ be as given in Definition 2 and let $\Xi(B_1, B_2, m_1, m_2, \delta)$ be as given in Definition 10. Consider $a, b \in (0, \frac{1}{2})$ with $b - a \geq \delta$.

Suppose that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} y_i \leq b \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{2} - a.$$

Then there exists a constant $\varepsilon_1 > 0$ (dependent only on ε_0 and δ), so that the following is true for large x : If $q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ and $q > x^{1/2-\varepsilon_1}$, then q has some $r \mid q$ with $r \in [x^a, x^b]$.

Proof. Recall that we defined $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ to be the following set:

$$\left\{ ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i \in [1, x] : \begin{array}{ll} \log_x(f_1 \dots f_m) \leq (1 - \varepsilon_0)B_{j,m}, & \log_x(f'_1 \dots f'_{m'}) \leq (1 - \varepsilon_0)B_{j',m'}, \\ \log_x(f_1 \dots f_m e) \leq (1 - \varepsilon_0)(A_j - \varepsilon), & \log_x(f'_1 \dots f'_{m'} e') \leq (1 - \varepsilon_0)(A_{j'} + \varepsilon), \\ ee' \text{ is } x^\delta\text{-smooth,} & \log_x(f_i), \log_x(f'_i) \geq \delta \text{ for all } i \end{array} \right\}$$

Fix some $q = ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ with $q > x^{1/2-\epsilon}$, where ϵ is very small. Let $p_1 \dots p_s$ be the prime factorization of ee' . We then know that $p_i < x^\delta$ for all i . We set

$$\begin{aligned} y_i &= \frac{1}{(1 - \varepsilon_0)} \log_x(f_i) & \text{for } 1 \leq i \leq m, \\ y_{m+i} &= \frac{1}{(1 - \varepsilon_0)} \log_x(f'_i) & \text{for } 1 \leq i \leq m', \\ z_i &= \log_x(p_i) & \text{for } 1 \leq i \leq s. \end{aligned}$$

Since $y_i \geq \delta$ for all i and $\sum_{i=1}^m y_i \leq B_{j,m}$ and $\sum_{i=m+1}^{m+m'} y_i \leq B_{j',m'}$, there exists a partition $I_1 \cup I_2$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} (1 - \varepsilon_0) y_i \leq (1 - \varepsilon_0) b < b \quad \text{and} \quad \sum_{i \in I_2} (1 - \varepsilon_0) y_i \leq (1 - \varepsilon_0) \left(\frac{1}{2} - a \right).$$

Since $q > x^{1/2-\epsilon}$, we also know that

$$\sum_{i \in I_1} (1 - \varepsilon_0) y_i + \sum_{i=1}^s z_i \geq \frac{1}{2} - \epsilon - (1 - \varepsilon_0) \left(\frac{1}{2} - a \right) \geq a,$$

provided ϵ is sufficiently small compared to ε_0 . But $z_i < \delta$ and so there exists $J \subseteq \{1, \dots, s\}$ with

$$\sum_{i \in I_1} (1 - \varepsilon_0) y_i + \sum_{i \in J} z_i \in [a, b].$$

Multiplying the corresponding f_i , f'_i and p_i together, we then get a factor $r \mid q$ with $r \in [x^a, x^b]$. $\square$

4.2.2. *3 factors.* Similarly, we next ask what condition ensures that $q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ has a factorization $q = uvr$ with $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$.

Lemma 12. *Let $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ be as given in Definition 2 and let $\Xi(B_1, B_2, m_1, m_2, \delta)$ be as given in Definition 10. Consider $a_1, a_2, b_1, b_2 \in (0, \frac{1}{2})$ with $b_1 - b_2 \geq a_1 - a_2$ and $b_1 + b_2 < \frac{1}{2}$ and $b_1 - a_1 \geq \delta$ and $b_2 - a_2 \geq \delta$.*

Suppose that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2 \cup I_3$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} y_i \leq b_1 \quad \text{and} \quad \sum_{i \in I_2} y_i \leq b_2 \quad \text{and} \quad \sum_{i \in I_3} y_i \leq \frac{1}{2} - b_1 - a_2.$$

Then there exists a constant $\varepsilon_1 > 0$ (dependent only on ε_0 and δ), so that the following is true for large x : If $q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ and $q > x^{1/2-\varepsilon_1}$, then q has a factorization $q = uvr$ with $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$.

Proof. The proof of Lemma 12 is very similar to the proof of Lemma 11:

We first fix some $q = ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ with $q > x^{1/2-\epsilon}$, where ϵ is very small. Denoting by $p_1 \dots p_s$ the prime factorization of ee' , we know that $p_i < x^\delta$ for all i and set

$$\begin{aligned} y_i &= \frac{1}{(1-\varepsilon_0)} \log_x(f_i) & \text{for } 1 \leq i \leq m, \\ y_{m+i} &= \frac{1}{(1-\varepsilon_0)} \log_x(f'_i) & \text{for } 1 \leq i \leq m', \\ z_i &= \log_x(p_i) & \text{for } 1 \leq i \leq s. \end{aligned}$$

Then there exists a partition $I_1 \cup I_2 \cup I_3$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} (1-\varepsilon_0)y_i < b_1 \quad \text{and} \quad \sum_{i \in I_2} (1-\varepsilon_0)y_i < b_2 \quad \text{and} \quad \sum_{i \in I_3} (1-\varepsilon_0)y_i \leq (1-\varepsilon_0) \left(\frac{1}{2} - b_1 - a_2 \right).$$

Since $q > x^{1/2-\epsilon}$, we also know that

$$\sum_{i \in I_2} (1-\varepsilon_0)y_i + \sum_{i=1}^s z_i > \frac{1}{2} - \epsilon - b_1 - (1-\varepsilon_0) \left(\frac{1}{2} - b_1 - a_2 \right) \geq a_2.$$

But $z_i < \delta$ for all i and so there exists $J_2 \subseteq \{1, \dots, s\}$ with

$$\sum_{i \in I_2} (1-\varepsilon_0)y_i + \sum_{i \in J_2} z_i \in [a_2, b_2].$$

Further, we then also have

$$\sum_{i \in I_1} (1-\varepsilon_0)y_i + \sum_{i \notin J_2} z_i > \frac{1}{2} - \epsilon - b_2 - (1-\varepsilon_0) \left(\frac{1}{2} - b_1 - a_2 \right) \geq b_1 - b_2 + a_2 \geq a_1.$$

Hence there also exists $J_1 \subseteq \{1, \dots, s\} \setminus J_2$ with

$$\sum_{i \in I_1} (1-\varepsilon_0)y_i + \sum_{i \in J_1} z_i \in [a_1, b_1].$$

Multiplying the corresponding f_i , f'_i and p_i together, we then get a factorization $q = uvr$ with $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$. $\square$

4.2.3. *4 factors.* Finally, we need a condition which ensures that $q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ has a factorization $q = uvr$ with the following properties: $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$ and there exists $d_1 \mid r$ with $d_1 \in [r^2 x^{a_3}, r^2 x^{b_3}]$. This is the condition which is typically the hardest to satisfy, and so we work more carefully and consider moduli q of a specific size, given by $q \asymp x^{1/2+2\omega_0}$.

Lemma 13. *Let $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ be as given in Definition 2 and let $\Xi(B_1, B_2, m_1, m_2, \delta)$ be as given in Definition 10. Consider $a_1, a_2, a_3, b_1, b_2, b_3$ with $3(b_1 - a_1) + (a_3 - b_3) \geq 0$ and $b_1 - b_2 \geq a_1 - a_2$ and $b_1 - a_1 \geq \delta$ and $b_2 - a_2 \geq \delta$ and $b_3 - a_3 \geq \delta$.*

Let $\omega_0 \geq -\varepsilon_0$ and suppose that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2 \cup I_3 \cup I_4$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} y_i \leq 2a_1 + b_3 \quad \text{and} \quad \sum_{i \in I_2} y_i \leq b_2 \quad \text{and} \quad \sum_{i \in I_3} y_i \leq \frac{1}{2} + 2\omega_0 - b_1 - a_2 \quad \text{and} \quad \sum_{i \in I_4} y_i \leq a_1 - 2b_1 - a_3.$$

Then for large x , every $q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ with $q \asymp x^{1/2+2\omega_0}$ has a factorization $q = uvr$ with the following properties: $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$ and there exists $d_1 \mid r$ with $d_1 \in [r^2 x^{a_3}, r^2 x^{b_3}]$.

Proof. We first fix some $q = ee' \prod_{i=1}^m f_i \prod_{i=1}^{m'} f'_i \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ with $q \asymp x^{1/2+2\omega_0}$. Denoting by $p_1 \dots p_s$ the prime factorization of ee' , we know that $p_i < x^\delta$ and set

$$\begin{aligned} y_i &= \log_x(f_i) & \text{for } 1 \leq i \leq m, \\ y_{m+i} &= \log_x(f'_i) & \text{for } 1 \leq i \leq m', \\ z_i &= \log_x(p_i) & \text{for } 1 \leq i \leq s. \end{aligned}$$

Then there exists a partition $I_1 \cup I_2 \cup I_3 \cup I_4$ of $\{1, \dots, m + m'\}$ with

$$\sum_{i \in I_1} y_i \leq 2a_1 + b_3 \quad \text{and} \quad \sum_{i \in I_2} y_i \leq b_2 \quad \text{and} \quad \sum_{i \in I_3} y_i \leq \frac{1}{2} + 2\omega_0 - b_1 - a_2 \quad \text{and} \quad \sum_{i \in I_4} y_i \leq a_1 - 2b_1 - a_3.$$

Since $q \asymp x^{1/2+2\omega_0}$, we also know that

$$\begin{aligned} \sum_{i \in I_2} y_i + \sum_{i=1}^s z_i &> \frac{1}{2} + 2\omega_0 - (2a_1 + b_3) - \left(\frac{1}{2} + 2\omega_0 - b_1 - a_2\right) - (a_1 - 2b_1 - a_3) \\ &\geq a_2 + (3b_1 - 3a_1 + a_3 - b_3) \geq a_2. \end{aligned}$$

But $z_i < \delta$ for all i and so there exists $J_2 \subseteq \{1, \dots, s\}$ with

$$\sum_{i \in I_2} y_i + \sum_{i \in J_2} z_i \in [a_2, b_2].$$

This sum corresponds to factor u . Next, we observe that

$$\begin{aligned} \sum_{i \in I_1} y_i + \sum_{i \in I_4} y_i &\leq (2a_1 + b_3) + (a_1 - 2b_1 - a_3) = b_1 + (3a_1 - 3b_1 + b_3 - a_3) \leq b_1, \\ \sum_{i \in I_1} y_i + \sum_{i \in I_4} y_i + \sum_{i \notin J_2} z_i &> \frac{1}{2} + 2\omega_0 - b_2 - \left(\frac{1}{2} + 2\omega_0 - b_1 - a_2\right) \geq b_1 - b_2 + a_2 \geq a_1. \end{aligned}$$

Hence there also exists $J_1 \subseteq \{1, \dots, s\} \setminus J_2$ with

$$\sum_{i \in I_1} y_i + \sum_{i \in I_4} y_i + \sum_{i \in J_1} z_i \in [a_1, b_1].$$

This sum corresponds to factor r . Finally, consider $c \in [a_1, b_1]$. Notice that

$$\begin{aligned} \sum_{i \in I_1} y_i &\leq 2a_1 + b_3 \leq 2c + b_3, \\ \sum_{i \in I_1} y_i + \sum_{i \in J_1} z_i &\geq a_1 - (a_1 - 2b_1 - a_3) \geq 2b_1 + a_3 \geq 2c + a_3. \end{aligned}$$

Hence for any $c \in [a_1, b_1]$ we can find $J' \subseteq J_1$ with

$$\sum_{i \in I_1} y_i + \sum_{i \in J'} z_i \in [2c + a_3, 2c + b_3].$$

This sum corresponds to factor d_1 of r . Overall, multiplying the corresponding f_i , f'_i and p_i together, we then get that $q \asymp x^{1/2+2\omega_0}$ has a factorization $q = uvr$ with $r \in [x^{a_1}, x^{b_1}]$ and $u \in [x^{a_2}, x^{b_2}]$ so that for every $R \in [x^{a_1}, x^{b_1}]$, there exists $d_1 \mid r$ with $d_1 \in [R^2 x^{a_3}, R^2 x^{b_3}]$. $\square$

4.3. An equidistribution criterion. Recall now that our goal is to find conditions on δ , $\underline{A}$ and $\underline{B}$ which ensure that every $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$ has equidistribution properties for moduli in $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$. Combining the partition lemmas we just proved with the equidistribution estimates of Section 3, we can rephrase these conditions in terms of properties of $\Xi(B_1, B_2, m_1, m_2, \delta)$, as seen in Proposition 3.

Proposition 3. *Let $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ be as given in Definition 1, let $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ be as given in Definition 2 and let $\Xi(B_1, B_2, m_1, m_2, \delta)$ be as given in Definition 10.*

Let $\epsilon = 10^{-10}$. Suppose $\xi_1, \xi_2, \xi_3 \in (0, 1)$ satisfy the following conditions:

- (I) *Type I condition: $\min\{\xi_1 - 4A_n + \frac{2}{3}, \frac{9}{7} - \frac{34}{7}A_n\} - 2\epsilon > \delta$. (Recall that $\underline{A} = (A_1, \dots, A_n)$.)*
- (II) *Type II condition: $\frac{19}{2} - 36A_n - 13\delta + 100\epsilon \geq 0$ and*

$$\min\left\{\frac{\xi_2}{10} - \frac{32A_n}{10} + \frac{8}{10}, \frac{\xi_2}{4} + \frac{11}{16} - 3A_n\right\} - 2\epsilon \geq \delta.$$

(III) *Type III condition:* $\frac{11}{8} - \frac{7}{2}A_n - \frac{9}{8}\xi_3 - 2\epsilon > \delta$.

Suppose further that for all $j, j' \in \{1, \dots, n\}$ and for all $m, m' \in \{1, \dots, \lfloor \frac{1}{\delta} \rfloor\}$ with $m + m' > 0$, every tuple $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ has all of the following properties:

(A) *Type I condition:* There exists a partition $I_1 \cup I_2$ of $\{1, \dots, m + m'\}$ with

$$\sum_{i \in I_1} y_i \leq \xi_1 - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{6} - 4\omega(j, j') - 2\epsilon \quad \text{where} \quad \omega(j, j') = \left(\frac{A_j + A_{j'}}{2} - \frac{1}{4} \right).$$

(B) *Type IIa condition:* There exists a partition $I_1 \cup I_2$ of $\{1, \dots, m + m'\}$ with

$$\sum_{i \in I_1} y_i \leq \frac{2}{5} + \frac{24\omega(j, j')}{5} + \frac{7\delta}{5} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{14} - \frac{24\omega(j, j')}{7} - 2\epsilon.$$

(C) *Type IIb condition:* There also exists a partition $I_1 \cup I_2 \cup I_3$ of $\{1, \dots, m + m'\}$ with

$$\begin{aligned} \sum_{i \in I_1} y_i &\leq \frac{1}{3} + \frac{24\omega(j, j')}{3} + \frac{7\delta}{3} - 4\epsilon, \\ \sum_{i \in I_2} y_i &\leq \frac{1}{10} - \frac{34\omega(j, j')}{5} - \frac{7\delta}{5} - 4\epsilon, \\ \sum_{i \in I_3} y_i &\leq \frac{1}{35} + \frac{22\omega(j, j')}{35} + \frac{21\delta}{35} - 4\epsilon. \end{aligned}$$

(D) *Type IIc condition:* For any $\omega_0 \in [-\epsilon, \omega(j, j')]$ and any γ with $\xi_2 - \epsilon \leq \gamma \leq \frac{1}{3} + \frac{24\omega(j, j')}{3} + \frac{7\delta}{3} + 3\epsilon$, there exists a partition $I_1 \cup I_2 \cup I_3 \cup I_4$ with

$$\begin{aligned} \sum_{i \in I_1} y_i &\leq \gamma - 2\delta - 8\omega_0 - \epsilon, \\ \sum_{i \in I_2} y_i &\leq \frac{1}{2} - \gamma - 2\omega_0 - \epsilon, \\ \sum_{i \in I_3} y_i &\leq 4\omega_0 + \delta - \epsilon, \\ \sum_{i \in I_4} y_i &\leq 8\omega_0. \end{aligned}$$

(E) *Type III condition:* There also exists a partition $I_1 \cup I_2$ of $\{1, \dots, m + m'\}$ with

$$\sum_{i \in I_1} y_i \leq 1 - 6\omega(j, j') - \frac{3\xi_3}{2} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{5\omega(j, j')}{2} + \frac{3\xi_3}{8} - 2\epsilon.$$

Then for every $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$, $x > 1$, $\varepsilon_0 > 0$, $A > 0$ and $a \in \mathbb{Z}$ with $(a, P(x)) = 1$,

$$\sum_{\substack{q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0) \\ q \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(q)}} f(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x, 2x] \\ (n, q) = 1}} f(n; x) \right| \ll_{A, \varepsilon_0} \frac{x}{\log(x)^A}.$$

Proof. Recall that $\mathcal{H}(\xi_1, \xi_2, \xi_3)$ consists of convolutions of three different types, I, II and III. Our goal is to show that for each type of f and for all $j, j' \in \{1, \dots, n\}$ and for all $m, m' \in \{1, \dots, \lfloor \frac{1}{\delta} \rfloor\}$,

$$\sum_{\substack{q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0) \\ q \text{ is squarefree}}} \left| \sum_{\substack{n \in [x, 2x] \\ n \equiv a(q)}} f(n; x) - \frac{1}{\phi(q)} \sum_{\substack{n \in [x, 2x] \\ (n, q) = 1}} f(n; x) \right| \ll_{A, \varepsilon_0} \frac{x}{\log(x)^A}.$$

We fix some choice of $j, j' \in \{1, \dots, n\}$ and $m, m' \in \{1, \dots, \lfloor \frac{1}{\delta} \rfloor\}$ and may assume $m + m' > 0$ (as the $m + m' = 0$ case follows trivially from the other cases). For $q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ we have $q \leq x^{(1-\varepsilon_0)(A_j + A_{j'})}$ and so we will work with

$$\omega_{\max} = \frac{A_j + A_{j'}}{2} - \frac{1}{4}.$$

Case I: First consider $f(n; x) = (\alpha \star \beta)(n; x)$, where α is a coefficient sequence at scale M and β is a smooth coefficient sequence at scale N with $M(x)N(x) \asymp x$ and $N(x) \geq x^{\xi_1 - \epsilon}$.

Initially we assume $\gamma \leq \frac{1}{2}$ and define $\delta^*(\gamma) = \gamma - 4\omega_{\max} - \frac{1}{3} - \epsilon$. For $\gamma \geq \xi_1 - \epsilon$ we now have

$$3\gamma - 12\omega_{\max} - 3\delta^*(\gamma) > 1.$$

Also, the assumption $\xi_1 - 4A_n + \frac{2}{3} - 2\epsilon > \delta$ ensures that $\delta^*(\gamma) \geq \delta$.

Now we apply Lemma 5 with $(\omega_{\max}, \gamma, \delta^*(\gamma))$ in the place of (ω, γ, δ) . For sufficiently small ϵ' , any $f(n; x)$ of Type I has good equidistribution properties for moduli in $D_I(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$. But this means $Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0)$ also has the desired equidistribution properties, provided it is a subset of $D_I(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$. Recall that for $\gamma \leq \frac{1}{2}$,

$$D_I(x; \omega, \gamma, \delta; \epsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \exists r \mid d \text{ with } x^{\gamma-\delta-3\epsilon} < r < x^{\gamma-3\epsilon} \right\}.$$

To show $Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0) \subseteq D_I(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$, we now apply Lemma 11. We need that the relevant moduli q have a factor r with $x^{\gamma-\delta^*(\gamma)-3\epsilon'} < r < x^{\gamma-3\epsilon'}$ when $\xi_1 - \epsilon \leq \gamma \leq \frac{1}{2}$. But

$$\gamma - \delta^*(\gamma) - 3\epsilon' = 4\omega_{\max} + \frac{1}{3} + \epsilon - 3\epsilon'.$$

By Lemma 11 elements of $Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0)$ thus have the desired factors provided that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2$ of $\{1, \dots, m+m'\}$ with

$$\sum_{i \in I_1} y_i \leq \xi_1 - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{6} - 4\omega_{\max} - 2\epsilon.$$

On the other hand, if $\gamma \in (\frac{1}{2}, \frac{1}{2} + 2\omega_{\max} + \epsilon']$, we use $\delta^*(\gamma) = \frac{1}{14} - \frac{68}{14}\omega_{\max} - \epsilon$, so that

$$(4.5) \quad 68\omega_{\max} + 14\delta^*(\gamma) < 1.$$

Then $f(n; x)$ equidistributes for moduli in $D_I(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$. This time, D_I consists of q which have a factor r with $x^{1-\gamma-\delta^*(\gamma)-3\epsilon'} < r < x^{1-\gamma-3\epsilon'}$. Thus Lemma 11 applies to $Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0)$ provided that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2$ with

$$(4.6) \quad \sum_{i \in I_1} y_i \leq \frac{1}{2} - 2\omega_{\max} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{14} - \frac{68}{14}\omega_{\max} - 2\epsilon.$$

Case II: Next consider $f(n; x) = (\alpha \star \beta)(n; x)$, where α is coefficient sequence at scale M , β is a coefficient sequence at scale N with $M(x)N(x) \asymp x$, and $x^{\xi_2-\epsilon} \leq N(x) = x^\gamma \leq x^{1-\xi_2+\epsilon}$. Assume also that α and β both have the Siegel–Walfisz property. By symmetry, we only need to consider $N(x) \leq x^{1/2}$.

We will now apply the equidistribution estimates given in Lemma 3, Lemma 4 and Lemma 6.

Case IIa: We begin with Lemma 3 and set $\delta^*(\gamma) = \frac{5\gamma}{7} - \frac{2}{7} - \frac{24\omega_{\max}}{7} - \epsilon$. This choice ensures that

$$\begin{aligned} 24\omega_{\max} + 7\delta^*(\gamma) - 5\gamma &< -2, \\ 8\omega_{\max} + 3\delta^*(\gamma) - \gamma &< 0, \end{aligned}$$

so that we can apply Lemma 3 with $(\omega_{\max}, \gamma, \delta^*(\gamma))$ in the place of (ω, γ, δ) . We get that f has good equidistribution properties for moduli in $D_{IIa}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$, provided ϵ' is very small. Now we need to show that $Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0) \subseteq D_{IIa}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \epsilon')$. To apply Lemma 11, we need to restrict to γ with $\delta^*(\gamma) = \frac{5\gamma}{7} - \frac{2}{7} - \frac{24\omega_{\max}}{7} - \epsilon \geq \delta$. This means for now we only consider

$$\gamma \geq \frac{2}{5} + \frac{24\omega_{\max}}{5} + \frac{7\delta}{5} + 2\epsilon.$$

We then get that any $q \in Q(x; \delta, \underline{A}, \underline{B}, \epsilon, j, j', m, m', \epsilon_0)$ has a divisor $r \mid q$ with $x^{\gamma-3\epsilon-\delta^*(\gamma)} < r < x^{\gamma-3\epsilon}$ provided that for every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2$ with

$$(4.7) \quad \sum_{i \in I_1} y_i \leq \frac{2}{5} + \frac{24\omega_{\max}}{5} + \frac{7\delta}{5} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{14} - \frac{24\omega_{\max}}{7} - 2\epsilon.$$

Case IIb: Now we consider $\gamma < \frac{2}{5} + \frac{24\omega_{\max}}{5} + \frac{7\delta}{5} + 2\epsilon$ and use Lemma 4. Here we instead choose $\delta^*(\gamma) = \frac{3\gamma}{7} - \frac{1}{7} - \frac{24\omega_{\max}}{7} - \epsilon$, which ensures that

$$\begin{aligned} 24\omega_{\max} + 7\delta^*(\gamma) - 3\gamma &< -1, \\ 8\omega_{\max} + 3\delta^*(\gamma) - \gamma &< 0. \end{aligned}$$

Hence we can apply Lemma 4 with $(\omega_{\max}, \gamma, \delta^*(\gamma))$ and get that f has good equidistribution properties for moduli in $D_{IIb}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \varepsilon')$. To show $Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0) \subseteq D_{IIb}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \varepsilon')$, we apply Lemma 12. However, we need to restrict to γ with $\delta^*(\gamma) = \frac{3\gamma}{7} - \frac{1}{7} - \frac{24\omega_{\max}}{7} - \epsilon \geq \delta$, so have

$$\frac{1}{3} + \frac{24\omega_{\max}}{3} + \frac{7\delta}{3} + 3\epsilon < \gamma < \frac{2}{5} + \frac{24\omega_{\max}}{5} + \frac{7\delta}{5} + 2\epsilon$$

Recall now that

$$D_{IIb}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1/2-\gamma-2\omega-6\varepsilon-\delta} < u < x^{1/2-\gamma-2\omega-6\varepsilon} \end{array} \right\}.$$

Any $q \in Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0)$ has a divisor $r \mid q$ and a divisor $u \mid \frac{q}{r}$ with $x^{\gamma-3\varepsilon-\delta^*(\gamma)} < r < x^{\gamma-3\varepsilon}$ and $x^{1/2-\gamma-2\omega_{\max}-6\varepsilon-\delta^*(\gamma)} < u < x^{1/2-\gamma-2\omega_{\max}-6\varepsilon}$ provided that the following is true: For every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2 \cup I_3$ with

$$\sum_{i \in I_1} y_i \leq \gamma - 2\epsilon, \quad \sum_{i \in I_2} y_i \leq 1/2 - \gamma - 2\omega_{\max} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_3} y_i \leq \frac{3\gamma}{7} - \frac{1}{7} - \frac{10\omega_{\max}}{7} - 2\epsilon.$$

Of course we could simply check this condition for every γ in the relevant range, but Case IIb generally does not correspond to the most critical condition (Case IIc does), and so we simplify the above condition to the following criterion: There exists a partition $I_1 \cup I_2 \cup I_3$ with

$$(4.8) \quad \begin{aligned} \sum_{i \in I_1} y_i &\leq \frac{1}{3} + \frac{24\omega_{\max}}{3} + \frac{7\delta}{3} - 4\epsilon, \\ \sum_{i \in I_2} y_i &\leq \frac{1}{10} - \frac{34\omega_{\max}}{5} - \frac{7\delta}{5} - 4\epsilon, \\ \sum_{i \in I_3} y_i &\leq \frac{1}{35} + \frac{22\omega_{\max}}{35} + \frac{21\delta}{35} - 4\epsilon. \end{aligned}$$

Case IIc: Finally, we use Lemma 6 to treat the case $\xi_2 - \epsilon \leq \gamma \leq \frac{1}{3} + \frac{24\omega_{\max}}{3} + \frac{7\delta}{3} + 3\epsilon$. Here we will work with a function $\delta^*(\gamma)$ which satisfies

$$\delta^*(\gamma) \leq \min \left\{ \frac{1}{4} - 2\omega_{\max} - \frac{\gamma}{2}, \frac{\gamma}{10} - \frac{32\omega_{\max}}{10}, \frac{\gamma}{4} - \frac{1}{16} - 3\omega_{\max} \right\} - \epsilon.$$

This ensures that we have the following inequalities:

$$\begin{aligned} 8\omega_{\max} + 4\delta^*(\gamma) + 2\gamma &< 1, \\ 32\omega_{\max} + 10\delta^*(\gamma) - \gamma &< 0, \\ 48\omega_{\max} + 16\delta^*(\gamma) - 4\gamma &< -1. \end{aligned}$$

Hence we can apply Lemma 6 with $(\omega_{\max}, \gamma, \delta^*(\gamma))$ and get that f has good equidistribution properties for moduli in $D_{IIc}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \varepsilon')$. Recall here that

$$D_{IIc}(x; \omega, \gamma, \delta; \varepsilon) = \left\{ d \in [1, x^{1/2+2\omega}] \cap \mathbb{N} : \begin{array}{l} \exists r \mid d \exists u \mid \frac{d}{r} \exists d_1 \mid r \text{ with } x^{\gamma-3\varepsilon-\delta} < r < x^{\gamma-3\varepsilon} \\ \text{and } x^{1-\gamma-6\varepsilon-\delta} d^{-1} < u < x^{1-\gamma-6\varepsilon} d^{-1} \\ \text{and } r^2 x^{2-\gamma-52\varepsilon-\delta} d^{-4} < d_1 < r^2 x^{2-\gamma-52\varepsilon} d^{-4} \end{array} \right\}.$$

We now ask if the given upper bound on $\delta^*(\gamma)$ is sufficiently large to allow $\delta^*(\gamma) \geq \delta$ for $\xi_2 - \epsilon \leq \gamma \leq \frac{1}{3} + \frac{24\omega_{\max}}{3} + \frac{7\delta}{3} + 3\epsilon$. On the one hand, we need

$$\frac{1}{4} - 2\omega_{\max} - \frac{1}{2} \left(\frac{1}{3} + \frac{24\omega_{\max}}{3} + \frac{7\delta}{3} + 3\epsilon \right) \geq \delta,$$

which is implied by

$$\frac{19}{2} - 36A_n - 13\delta + 100\epsilon \geq 0.$$

On the other hand, we also need that

$$\min \left\{ \frac{\xi_2}{10} - \frac{32\omega_{\max}}{10}, \frac{\xi_2}{4} - \frac{1}{16} - 3\omega_{\max} \right\} - 2\epsilon \geq \delta.$$

Recalling that $\omega_{\max} \leq A_n - \frac{1}{4}$, it suffices to have

$$\min \left\{ \frac{\xi_2}{10} - \frac{32A_n}{10} + \frac{8}{10}, \frac{\xi_2}{4} + \frac{11}{16} - 3A_n \right\} - 2\epsilon \geq \delta.$$

These two criteria form condition (II) of Proposition 3. Hence we can work with any $\delta^*(\gamma)$ which satisfies

$$\delta \leq \delta^*(\gamma) \leq \min \left\{ \frac{1}{4} - 2\omega_{\max} - \frac{\gamma}{2}, \frac{\gamma}{10} - \frac{32\omega_{\max}}{10}, \frac{\gamma}{4} - \frac{1}{16} - 3\omega_{\max} \right\} - \epsilon.$$

Now we can apply Lemma 13 to show that $Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0) \subseteq D_{IIc}(x; \omega_{\max}, \gamma, \delta^*(\gamma); \varepsilon')$. According to Lemma 13, every $q \in Q^*(x; \delta, \underline{A}, \underline{B}, \varepsilon, \varepsilon_0)$ has a suitable factorization provided the following is true: For every $\omega_0 \in [-\varepsilon_0, \omega_{\max}]$ and every tuple $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ there exists a partition $I_1 \cup I_2 \cup I_3 \cup I_4$ of $\{1, \dots, m+m'\}$ which satisfies inequalities

$$\begin{aligned} \sum_{i \in I_1} y_i &\leq \gamma - 2\delta^*(\gamma) - 8\omega_0 - \epsilon, \\ \sum_{i \in I_2} y_i &\leq \frac{1}{2} - \gamma - 2\omega_0 - \epsilon, \\ \sum_{i \in I_3} y_i &\leq 4\omega_0 + \delta^*(\gamma) - \epsilon, \\ \sum_{i \in I_4} y_i &\leq 8\omega_0. \end{aligned}$$

Looking at these conditions, it is actually often better to work with a smaller δ . Since $\delta^*(\gamma) \geq \delta$, we hence settle on the following criterion:

$$\begin{aligned} \sum_{i \in I_1} y_i &\leq \gamma - 2\delta - 8\omega_0 - \epsilon, \\ \sum_{i \in I_2} y_i &\leq \frac{1}{2} - \gamma - 2\omega_0 - \epsilon, \\ \sum_{i \in I_3} y_i &\leq 4\omega_0 + \delta - \epsilon, \\ \sum_{i \in I_4} y_i &\leq 8\omega_0. \end{aligned}$$

Case III: Finally, consider $f(n; x) = (\alpha \star \psi_1 \star \psi_2 \star \psi_3)(n; x)$, where α is a coefficient sequence at scale M and ψ_1, ψ_2 and ψ_3 are smooth coefficient sequences at scales N_1, N_2 and N_3 with $M(x)N_1(x)N_2(x)N_3(x) \asymp x$. Additionally, $x^{1-2\xi_3-\epsilon} \leq N_1(x), N_2(x), N_3(x) \leq x^{\xi_3+\epsilon}$ and $N_i(x)N_j(x) \geq x^{1-\xi_3-\epsilon}$ for $i \neq j$. We define $\delta^* = \frac{1}{2} - \frac{7}{2}\omega_{\max} - \frac{9}{8}\xi_3 - \epsilon$, so that

$$28\omega_{\max} + 9\xi_3 + 8\delta^* < 4.$$

The condition $\frac{11}{8} - \frac{7}{2}A_n - \frac{9}{8}\xi_3 - 2\epsilon > \delta$ ensures $\delta^* > \delta$. Now we apply Lemma 7 with $(\omega_{\max}, \gamma, \delta^*)$ in the place of (ω, γ, δ) . We get that any $f(n; x)$ of Type III has good equidistribution properties for moduli in $D_{III}(x; \omega_{\max}, \gamma, \delta^*; \varepsilon')$. To obtain

$$Q(x; \delta, \underline{A}, \underline{B}, \varepsilon, j, j', m, m', \varepsilon_0) \subseteq D_{III}(x; \omega_{\max}, \gamma, \delta^*; \varepsilon')$$

we use once more Lemma 11. We need to show that elements q of Q have a factor $r \mid q$ with

$$x^{1/3+4\delta^*/3-4\omega_{\max}/3-\delta^*} < r < x^{1/3+4\delta^*/3-4\omega_{\max}/3}.$$

This is guaranteed if every $(y_1, \dots, y_{m+m'}) \in \Xi(B_{j,m}, B_{j',m'}, m, m', \delta)$ has a partition $I_1 \cup I_2$ with

$$\sum_{i \in I_1} y_i \leq \frac{1}{3} + \frac{4\delta^*}{3} - \frac{4\omega_{\max}}{3} \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{1}{6} - \frac{\delta^*}{3} + \frac{4\omega_{\max}}{3}.$$

Plugging in $\delta^* = \frac{1}{2} - \frac{7}{2}\omega_{\max} - \frac{9}{8}\xi_3 - \epsilon$, this is implied by

$$(4.9) \quad \sum_{i \in I_1} y_i \leq 1 - 6\omega_{\max} - \frac{3\xi_3}{2} - 2\epsilon \quad \text{and} \quad \sum_{i \in I_2} y_i \leq \frac{5\omega_{\max}}{2} + \frac{3\xi_3}{8} - 2\epsilon.$$

This concludes the proof. $\square$

The statement of Proposition 3 may look a bit complicated at first, but it is extremely useful for identifying suitable choices of $B_{j,m}$ for given $\underline{A}$ and δ . For instance, one may choose some small constant c and sample tuples in $\Xi(B_1, B_2, m_1, m_2, \delta)$ by considering $(y_1, \dots, y_{m_1+m_2})$ with $y_i \in c\mathbb{N}$. Using a computer, it is then easy to check if there is a partition $I_1 \cup I_2 \cup I_3 \cup I_4$ so that $(y_1, \dots, y_{m_1+m_2})$ satisfies $\sum_{i \in I_j} y_i < C_j$ for $j = 1, 2, 3, 4$. Choosing constants C_1, C_2, C_3 and C_4 according to the partitions listed in Proposition 3, but leaving room to spare to account for an error $(m_1 + m_2)c$, one can then find B_1 and B_2 with the desired partition properties. Of course, the smaller the choice of c , the longer this computation takes, but at least for small m and m' it is not difficult to find near optimal values for $B_{j,m}$ by using this approach.

However, to prove $H_1 \leq 240$, such a brute force computation is actually not necessary and we will instead pick $B_{j,m}$ for which the conditions of Proposition 3 can easily be verified by hand.

5. INTEGRALS VIA MATRIX MULTIPLICATION

In the previous sections we discussed the theoretical framework of our proof: we proved a variant of the GPY sieve for the support $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$, derived useful equidistribution lemmas and constructed a minorant for the prime indicator function. However, once we have chosen δ , $\underline{A}$, $\underline{B}$ and ε , one major obstacle remains: we need to solve the optimization problem described in Proposition 1 and find a function F which makes the following quantity large:

$$\frac{k(1 - c_1)J(F; \delta, \underline{A}, \underline{B}, \varepsilon) - kc_2K(F; \delta, \underline{A}, \underline{B}, \varepsilon)}{I(F; \delta, \underline{A}, \underline{B}, \varepsilon)} > 1.$$

5.1. A basis. To find a good function F , we first choose some symmetric functions $G_1, \dots, G_l$ as a basis and consider linear combinations $F = c_1G_1 + \dots + c_lG_l$. We set $\underline{c} = (c_1, \dots, c_l)$ and define matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ as follows:

$$\begin{aligned} (M_1)_{i,j} &= \int \cdots \int_{T_k(\delta, \underline{A}, \underline{B}, \varepsilon)} G_i(t_1, \dots, t_k) G_j(t_1, \dots, t_k) dt_1 \dots dt_k, \\ (M_2)_{i,j} &= k(1 - c_1) \sum_{m=1}^n \sum_{m'=1}^n \int \cdots \int_{\substack{t_1 + \dots + t_{i-1} + t_{i+1} + \dots + t_k \leq \max\{A_m - \varepsilon, A_{m'} - \varepsilon\} \\ t_1 + \dots + t_{i-1} + t_i + t_{i+1} + \dots + t_k \in [A_{m-1} + \varepsilon, A_m + \varepsilon] \\ t_1 + \dots + t_{i-1} + t'_i + t_{i+1} + \dots + t_k \in [A_{m'-1} + \varepsilon, A_{m'} + \varepsilon]}} G_i(t_1, \dots, t_{k-1}, t_k) G_j(t_1, \dots, t_{k-1}, t'_k) dt_k dt'_k dt_1 \dots dt_{k-1} \\ &\quad - kc_2 \sum_{m=1}^n \sum_{m'=1}^n \int \cdots \int_{\substack{t_1 + \dots + t_{k-1} > \max\{A_m - \varepsilon, A_{m'} - \varepsilon\} \\ t_1 + \dots + t_{k-1} + t_k \in [A_{m-1} + \varepsilon, A_m + \varepsilon] \\ t_1 + \dots + t_{k-1} + t'_k \in [A_{m'-1} + \varepsilon, A_{m'} + \varepsilon]}} G_i(t_1, \dots, t_k) G_j(t_1, \dots, t_k) dt_1 \dots dt_k \end{aligned}$$

This gives us the following identity:

$$(5.1) \quad \frac{\underline{c} \mathbf{M}_2 \underline{c}^T}{\underline{c} \mathbf{M}_1 \underline{c}^T} = \frac{k(1 - c_1)J(F; \delta, \underline{A}, \underline{B}, \varepsilon) - kc_2K(F; \delta, \underline{A}, \underline{B}, \varepsilon)}{I(F; \delta, \underline{A}, \underline{B}, \varepsilon)}.$$

To find the optimal choice for $\underline{c}$, Maynard [3] and Polymath [5] computed the largest eigenvalue of $\mathbf{M}_2 \mathbf{M}_1^{-1}$, and chose $\underline{c}$ to be a corresponding eigenvector. While our new choice of $\mathbf{M}_1$ and $\mathbf{M}_2$ may not be positive definite, we nonetheless follow the same approach: We compute exact values for the entries of $\mathbf{M}_1$ and $\mathbf{M}_2$, and then approximate numerically to obtain eigenvalues and eigenvectors. Choosing an eigenvector, we then approximate it via rational numbers, obtaining a good choice for $\underline{c}$. We then compute the ratio (5.1) exactly, and if the result is greater than 1, we have shown that $H_1 \leq H(k)$, where $H(k)$ is the length of the narrowest admissible k -tuple.

Regarding the choice of basis $G_1, \dots, G_l$, Polymath considered the set of symmetric polynomials

$$\mathcal{B}_D = \{p(t_1, \dots, t_k)^2(1 - t_1 - \dots - t_k)^b : p(t_1, \dots, t_k) \text{ symmetric polynomial, } \deg(p) = a, 2a + b \leq D\}.$$

In particular, $H_1 \leq 246$ was obtained by optimization over basis $\mathcal{B}_{27}$. To obtain $H_1 \leq 240$, we work with the smaller basis $\mathcal{B}_{19}$, which indicates that our method gives quite a large improvement.

5.2. Integration via recursion and matrix multiplication. To find eigenvectors of $\mathbf{M}_2 \mathbf{M}_1^{-1}$, one must of course first calculate the entries of matrices $\mathbf{M}_1$ and $\mathbf{M}_2$. We need to obtain exact integrals as even minor errors in the matrix entries quickly lead to incorrect results. However, trying to use Mathematica to integrate a polynomial in 49 variables over a region like $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ seems pretty hopeless: even for polynomials in 10 variables a computation can already take 10 minutes or more, and the duration appears to increase exponentially, so that we would never get any result for 49 variables. Therefore we need to find a better way to integrate polynomials over $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. We have carefully chosen the region to make it possible to completely rewrite integration in terms of recursion and matrix multiplication. Below we will briefly explain the key ideas behind this process.

5.2.1. *Building blocks.* We begin by considering much simpler sets

$$\begin{aligned} T_s(k) &= \{(t_1, \dots, t_k) \in [0, \delta]^k : t_1 + \dots + t_k \leq 1\}, \\ T_b(k) &= \{(t_1, \dots, t_k) \in [\delta, 1]^k : t_1 + \dots + t_k \leq 1\} \end{aligned}$$

and polynomials $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$. If one naively expands $(1 - t_1 - \dots - t_k)^b$ and integrates $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$ by hand, then the condition $t_1 + \dots + t_k \leq 1$ has the effect that each evaluation of $\int dt_i$ increases the polynomial's degree by 1. So we would quickly get an unmanageable number of different polynomials that need to be considered. To avoid such issues, we use the following two tricks:

First, we note that the integral of $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$ over $T_s(k)$ or $T_b(k)$ can itself be expressed (piecewise) as a polynomial in δ . Its degree is at most $k + a_1 + \dots + a_k + b$. The coefficients of this polynomial only depend on the value of $\lfloor \frac{1}{\delta} \rfloor$. We may write

$$\begin{aligned} \int_{T_s(k)} \dots \int_{T_s(k)} t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b dt_1 \dots dt_k &= \sum_{i=0}^{k+a_1+\dots+a_k+b} C_{m,i}(k, \underline{a}, b) \delta^i \quad \text{for } \left\lfloor \frac{1}{\delta} \right\rfloor \in [m, m+1], \\ \int_{T_b(k)} \dots \int_{T_b(k)} t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b dt_1 \dots dt_k &= \sum_{i=0}^{k+a_1+\dots+a_k+b} D_{m,i}(k, \underline{a}, b) \delta^i \quad \text{for } \left\lfloor \frac{1}{\delta} \right\rfloor \in [m, m+1]. \end{aligned}$$

The goal now is to find the coefficients $C_{m,i}(k, \underline{a}, b)$ and $D_{m,i}(k, \underline{a}, b)$.

Secondly, when $a_k \neq 0$ or $b \neq 0$ we introduce additional integrals to replace $t_k^{a_k}$ and $(1 - t_1 - \dots - t_k)^b$. This trick gives us a nice recursive definition for the coefficients $C_{m,i}(k, \underline{a}, b)$ and $D_{m,i}(k, \underline{a}, b)$. Here is a brief example:

$$\begin{aligned} \sum_{i=0}^{k+a_1+\dots+a_k+b} C_{m,i}(k, \underline{a}, b) \delta^i &= \int_0^\delta \dots \int_0^\delta t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b dt_1 \dots dt_k \\ &= \int_0^\delta \dots \int_0^\delta \int_0^{(1-t_1-\dots-t_k)^b} t_1^{a_1} \dots t_k^{a_k} ds dt_1 \dots dt_k \\ &= \int_0^1 (1 - s^{1/b})^{k+a_1+\dots+a_k} \int_0^{\frac{\delta}{1-s^{1/b}}} \dots \int_0^{\frac{\delta}{1-s^{1/b}}} u_1^{a_1} \dots u_k^{a_k} du_1 \dots du_k ds \\ &= \sum_{l=0}^{\infty} \int_{\max\{0, (1-\delta(l+1))^b\}}^{(1-\delta l)^b} (1 - s^{1/b})^{k+a_1+\dots+a_k} \sum_{i=0}^{k+a_1+\dots+a_k+b} C_{l,i}(k, \underline{a}, 0) \left(\frac{\delta}{1 - s^{1/b}} \right)^i ds. \end{aligned}$$

Evaluating the integral over s , we get another polynomial in δ . Carefully comparing coefficients on both sides, we can then express $C_{m,i}(k, \underline{a}, b)$ in terms of $C_{l,i}(k, \underline{a}, 0)$ for various l . Similarly, $C_{m,i}(k, \underline{a}, 0)$ can be expressed in terms of $C_{m,i}(k, (a_1, \dots, a_{k-1}), 0)$, $D_{m,i}(k, (a_1, \dots, a_{k-1}), 0)$, $C_{m-1,i}(k, (a_1, \dots, a_{k-1}), 0)$ and $D_{m-1,i}(k, (a_1, \dots, a_{k-1}), 0)$. This algorithm drastically speeds up integration over $T_s(k)$ for polynomials of the form $t_1^{a_1} \dots t_k^{a_k} (1 - t_1 - \dots - t_k)^b$.

5.2.2. *More complicated integrals.* The integrals discussed in the previous section work as building blocks for the calculation of more complicated integrals. For instance, we may now consider the region

$$T_{b,s}(k) = \{(u_1, \dots, u_r, v_1, \dots, v_s) \in [\delta, 1]^r \times [0, \delta]^s : u_1 + \dots + u_r + v_1 + \dots + v_s < A, v_1 + \dots + v_r < D\}.$$

Integration of $u_1^{a_1} \dots u_r^{a_r} v_1^{b_1} \dots v_s^{b_s} (A - u_1 - \dots - u_r - v_1 - \dots - v_s)^c$ over $T_{b,s}(k)$ can be expressed in terms of the coefficients $C_{m,i}(r, \underline{b}, n)$ and $D_{m,i}(s, \underline{a}, n)$:

$$\begin{aligned}
& \int_{T_{b,s}(k)} \dots \int u_1^{a_1} \dots u_r^{a_r} v_1^{b_1} \dots v_s^{b_s} (A - u_1 - \dots - u_r - v_1 - \dots - v_s)^c du_1 \dots du_r dv_1 \dots dv_s \\
&= \int_{\delta}^1 \dots \int_{\delta}^1 u_1^{a_1} \dots u_r^{a_r} \int_0^{\delta} \dots \int_0^{\delta} v_1^{b_1} \dots v_s^{b_s} (A - u_1 - \dots - u_r - v_1 - \dots - v_s)^c \\
&\quad A-D < u_1 + \dots + u_r < A \quad v_1 + \dots + v_s < A - u_1 - \dots - u_r \\
&+ \int_{\delta}^1 \dots \int_{\delta}^1 u_1^{a_1} \dots u_r^{a_r} \sum_{n=0}^c \binom{c}{n} (A - D - u_1 - \dots - u_r)^{c-n} \int_0^{\delta} \dots \int_0^{\delta} v_1^{b_1} \dots v_s^{b_s} (D - v_1 - \dots - v_s)^n \\
&\quad u_1 + \dots + u_r \leq A - D \quad v_1 + \dots + v_s < D \\
&= \sum_{\substack{l=0 \\ \max\{A-D, A-\delta l\} < u_1 + \dots + u_r < A-\delta(l+1)}}^{\infty} \int_{\delta}^1 \dots \int_{\delta}^1 u_1^{a_1} \dots u_r^{a_r} (A - u_1 - \dots - u_r)^{s+b_1+\dots+b_s+c} \sum_{i=0}^{s+b_1+\dots+b_s+c} C_{l,i}(s, \underline{b}, c) \left(\frac{\delta}{A - u_1 - \dots - u_r} \right)^i \\
&+ \int_{\delta}^1 \dots \int_{\delta}^1 u_1^{a_1} \dots u_r^{a_r} \sum_{n=0}^c \binom{c}{n} (A - D - u_1 - \dots - u_r)^{c-n} D^{s+b_1+\dots+b_s+n} \sum_{i=0}^{s+b_1+\dots+b_s+n} C_{\lfloor \frac{D}{\delta} \rfloor, i}(s, \underline{b}, c) \left(\frac{\delta}{D} \right)^i.
\end{aligned}$$

The computation above already replaced the integrals over $v_i < \delta$ by coefficients $C_{m,i}(s, \underline{b}, n)$, and the remaining integrals over $u_i > \delta$ can similarly be replaced by $D_{m,i}(r, \underline{a}, n)$. Overall, the entire integral can be expressed by multiplying a suitable matrix left and right by the coefficient vectors C and D .

The integral over $T_{b,s}(k)$ is itself a building block in our ultimate goal of integrating symmetric polynomials over $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$. My code decomposes all such integrals into matrix multiplications and recursions. The above discussion highlighted the key ideas behind my calculation process, but I will also upload my full code at a later time, when I have had time to insert more comments to make it reader friendly.

6. FINAL STEPS

To conclude the proof of $H_1 \leq 246$, we first briefly discuss our general strategy for proving bounds on H_1 via Proposition 1, Proposition 2 and Proposition 3.

6.1. General Strategy. Recall that $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ is defined as follows:

$$T_k(\delta, \underline{A}, \underline{B}, \varepsilon) = \bigcup_{j=1}^n \left\{ (t_1, \dots, t_k) \in [0, 1]^k : \sum_{i=1}^k t_i \in [A_{j-1} + \varepsilon, A_j + \varepsilon] \text{ and } \sum_{i \in I} t_i \leq B_{j, |I|} \text{ for } I = \{i : t_i > \delta\} \right\}.$$

We need to find good $\underline{A}$, $\underline{B}$, ε and δ . We start by selecting some vector $(A_1, \dots, A_n)$ and some $\delta > 0$. We then take a look at Proposition 3 and choose parameters (ξ_1, ξ_2, ξ_3) as follows:

$$\begin{aligned}
\xi_1 &\approx 4A_n + \delta - \frac{2}{3}, \\
\xi_2 &\approx \max \left\{ \frac{4}{10}, 32A_n - 8 + 10\delta, 12A_n - \frac{11}{4} + 4\delta \right\}, \\
\xi_3 &\approx \frac{11}{9} - \frac{28}{9}A_n - \frac{8}{9}\delta.
\end{aligned}$$

Next we apply Proposition 2 to construct a minorant $\rho(n; x)$ for the prime indicator function. If the conditions on ξ_1, ξ_2, ξ_3 stated in Proposition 2 fail, or if c_1 is large, we need to decrease the size of δ .

Having chosen (ξ_1, ξ_2, ξ_3) and $\underline{A}$ and δ , we then use Proposition 3 to find a suitable choice for $\underline{B}$, ideally with $B_{m,j}$ large. This determines the support $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ and we can now compute matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ and search for a good choice of vector $\underline{c}$. If we find $\underline{c}$ with

$$\frac{\underline{c} \mathbf{M}_2 \underline{c}^T}{\underline{c} \mathbf{M}_1 \underline{c}^T} > 1,$$

then Proposition 1 tells us that $H_1 \leq H(k)$.

6.2. Proof of Theorem 1. Proposition 1, Proposition 2 and Proposition 3 were all stated much more generally than is required for the proof of $H_1 \leq 240$. As will be discussed below, a quite simple choice of $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ can be used to get this result, and we can work directly with $1_{\mathbb{P}}(n)$ rather than a minorant $\rho(n; x)$. We do not need the full strength of Proposition 3 either. I developed these more general results with the intend of proving better bounds on H_1 , but due to time limitations, this paper only proves the minimal possible improvement.

Proof of Theorem 1. We choose $\varepsilon = 0.0075$ and $\underline{A} = (-\varepsilon, 0.253)$ and $\delta = 0.028$. Furthermore, we take $\underline{B} = (B_{1,1}, \dots, B_{1, \lfloor \frac{1}{\delta} \rfloor})$ with $B_{1,1} = B_{1,2} = 0.15$ and $B_{1,m} = 0.17$ for $m \geq 3$.

Next we choose $\xi_1 = 0.38$ and $\xi_2 = \xi_3 = 0.4$. For this choice, conditions (I), (II) and (III) of Proposition 3 are satisfied. Furthermore, the upper bound on the sum over I_1 in each of the conditions (A), (B), (C) and (E) is greater than 0.34. Since $B_{1,m} + B_{1,m'} \leq 0.34$, this means we can always simply choose $I_1 = \{1, \dots, m + m'\}$. In condition (D) we are done if we have $\{1, \dots, m + m'\} = I_1 \cup I_2$ with bounds

$$\sum_{i \in I_1} y_i \leq 0.32 \quad \text{and} \quad \sum_{i \in I_2} y_i \leq 0.07.$$

This is possible since $B_{1,m} + B_{1,m'} \leq 0.32$ if $m \leq 2$ or $m' \leq 2$, and for tuples with $m \geq 3$ we must have a subsum between 0.02 and 0.07. Thus by Proposition 3, $f \in \mathcal{H}(\xi_1, \xi_2, \xi_3)$ have the desired equidistribution properties. We also see that (ξ_1, ξ_2, ξ_3) satisfies all inequalities given in Proposition 2, and this proposition tells us to use $\rho(n; x) = 1_{\mathbb{P}}(n)$, which means $c_1 = c_2 = 0$. We compute the relevant matrices $\mathbf{M}_1$ and $\mathbf{M}_2$ for $k = 49$ and are able to obtain a vector $\underline{c}$ with

$$\frac{\underline{c} \mathbf{M}_2 \underline{c}^T}{\underline{c} \mathbf{M}_1 \underline{c}^T} > 1.$$

Therefore Proposition 1 tells us that $H_1 \leq 240$. □

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